

Bioquímica Estrutural:

Estudo da estrutura das moléculas biológicas e dos princípios que a regem.

(~ **Biologia Estrutural**)

Macromoléculas biológicas

- **DNA:** repositório da informação genética na maioria dos organismos vivos
- **RNA:** transferência (e repositório) de informação genética, matriz para a síntese proteica, funções estruturais, etc...
- **Proteínas:** componentes estruturais (pele, ossos, músculo, cabelo, etc...), catálise de reacções bioquímicas (enzimas), transmissão de sinais, regulação, transdução de energia, etc.,etc., etc.!..
- **Lípidos:** componentes essenciais das membranas biológicas, sinalização
- **Polissacáridos:** armazenamento de energia, função estrutural

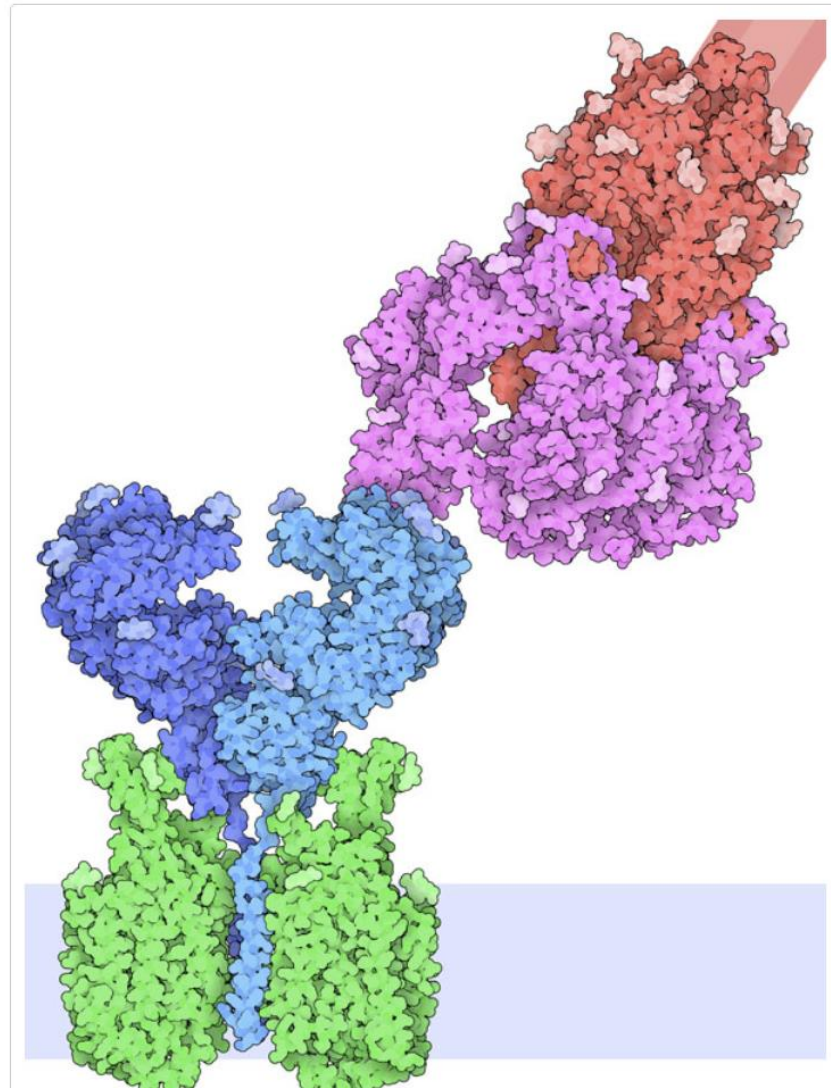
Sequência



Estrutura

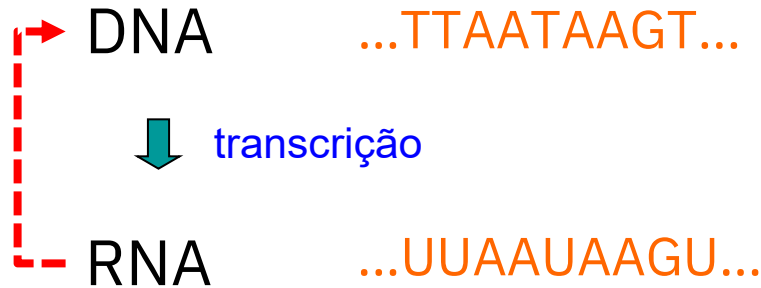


Função



Fluxo de informação biológica

vírus de RNA

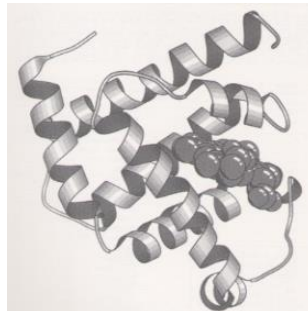


cadeia
polipeptídica ...LISVHDN...



priões

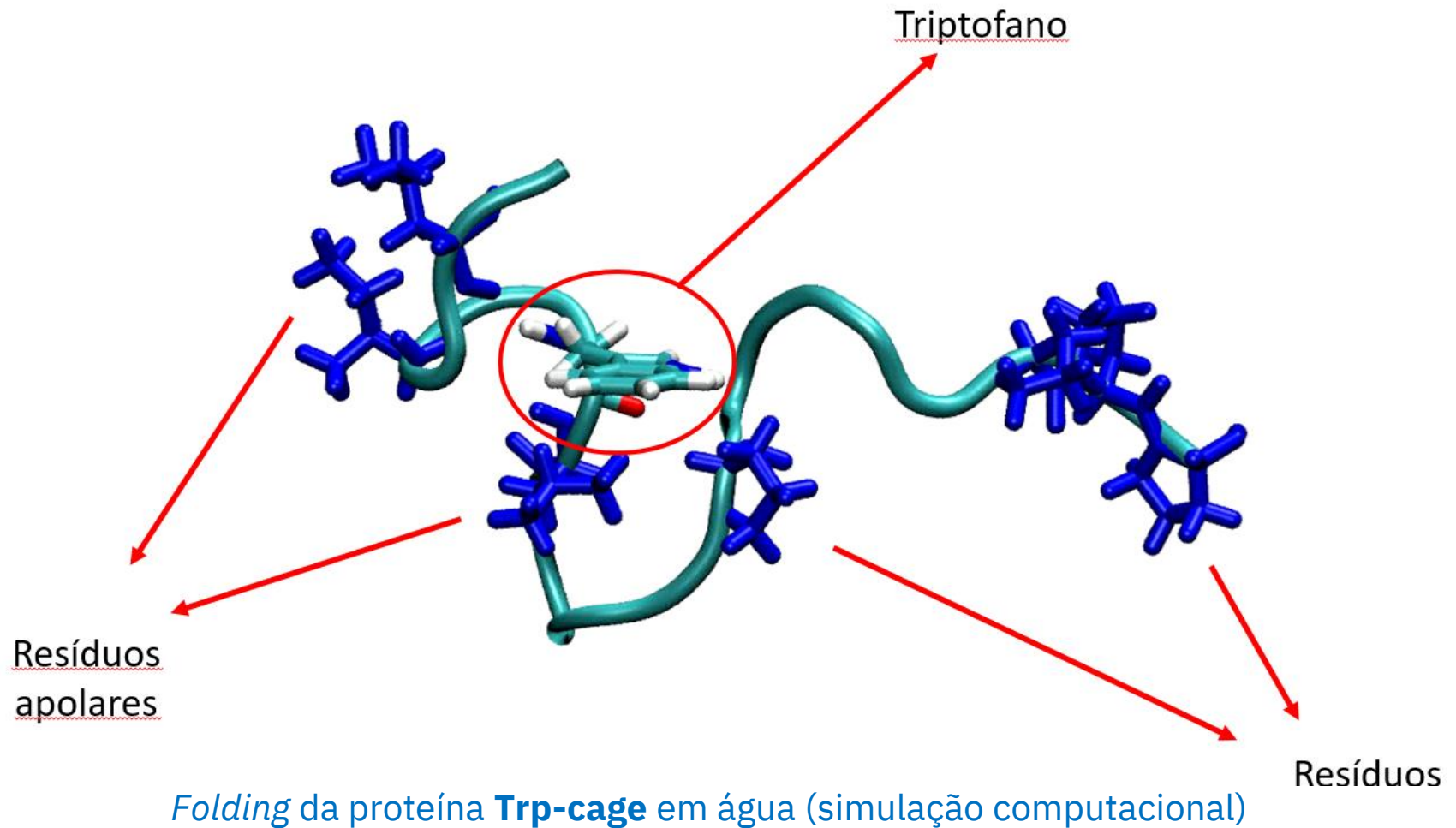
2025-2026



Dogma central da
biologia molecular

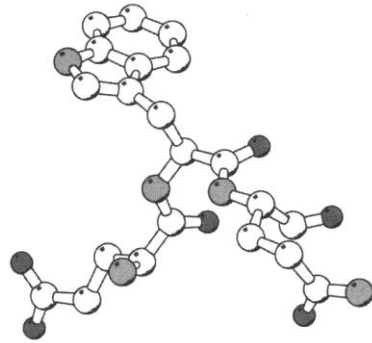
Excepções: vírus de RNA,
priões, ribozimas (?)

Aquisição espontânea da estrutura tridimensional (Sequência → Estrutura)

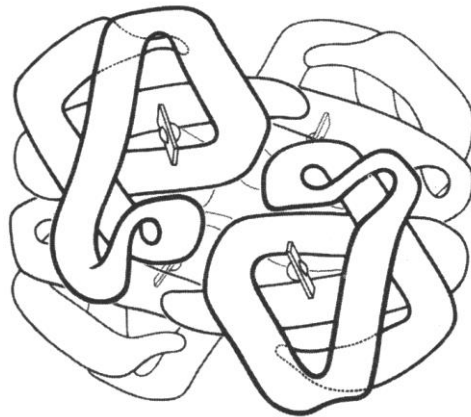
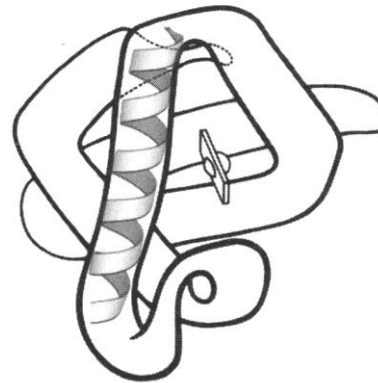
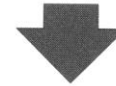
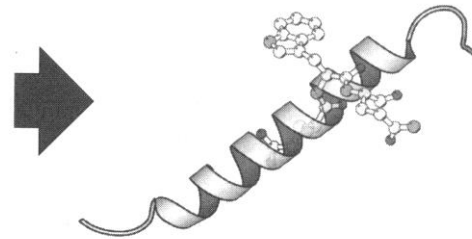


Níveis de organização da estrutura das proteínas

Estrutura primária



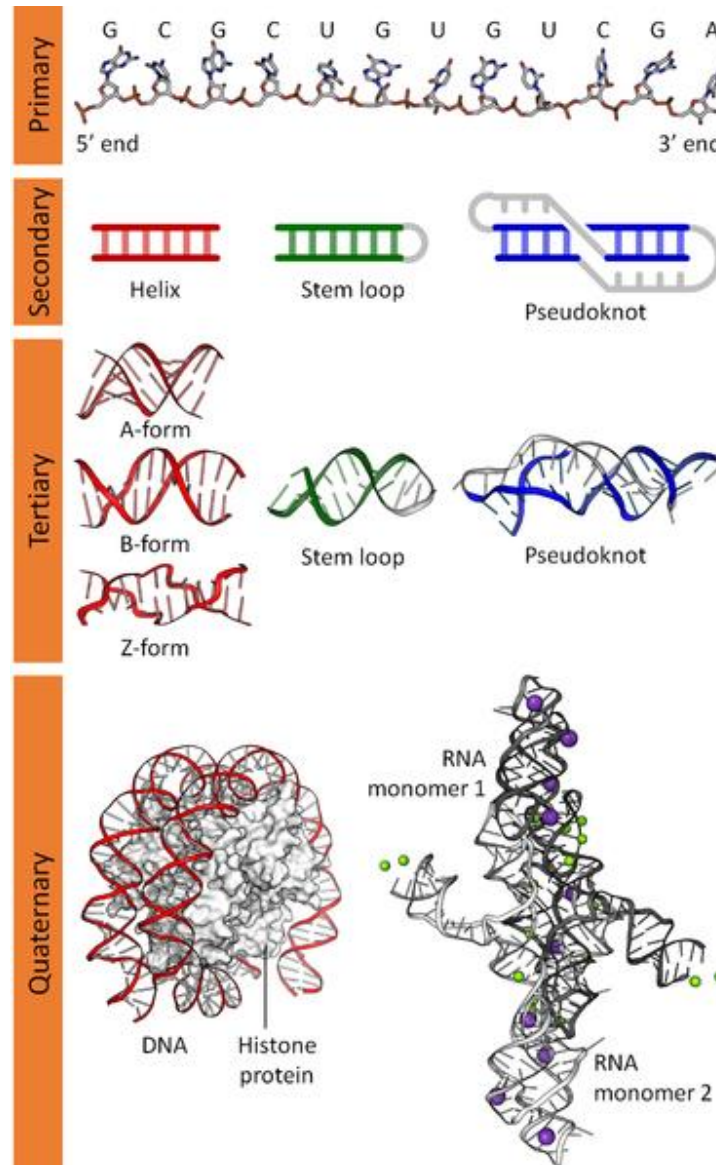
Estrutura secundária



Estrutura quaternária

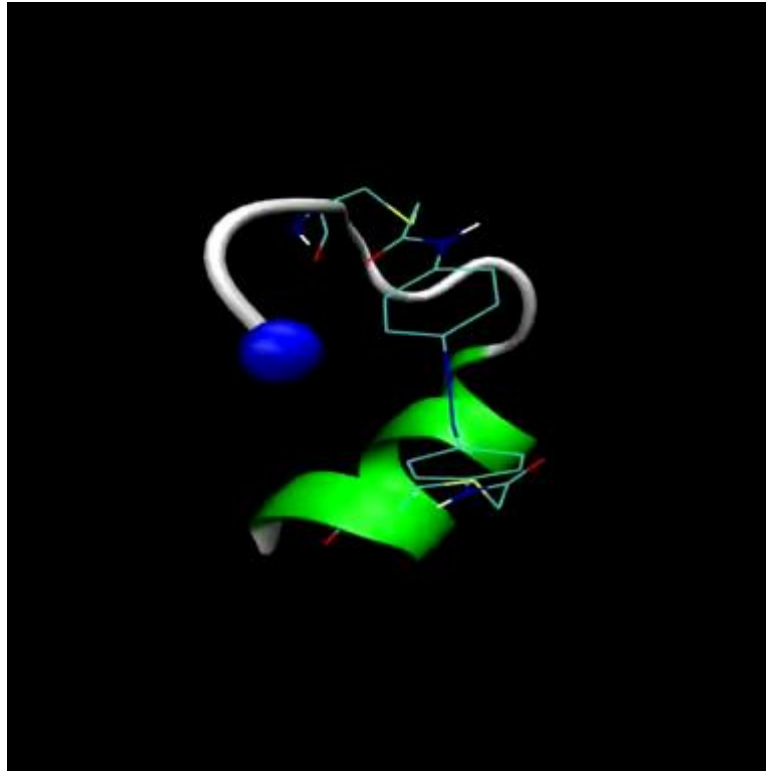
Estrutura terciária

Níveis de organização da estrutura de ácidos nucleicos



Estrutura quaternária
pode envolver moléculas
de proteína

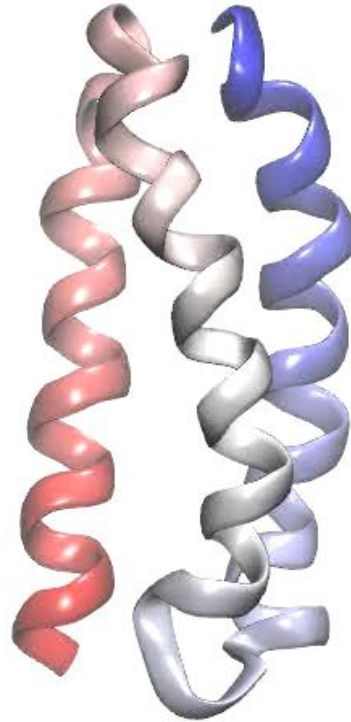
Formação hierárquica da estrutura



Formação espontânea de uma hélice α
(simulação)

Formação hierárquica da estrutura na proteína $\alpha 3D$

Three helix bundle: A3D



Estado desnaturado $\rightarrow$ Hélices $\rightarrow$ *Fold* Nativo
(simulação)

PDB: 2A3D

De onde provem a informação estrutural ?

Combinação de vários tipos de conhecimento:

- Teoria da ligação química
- Geometria de moléculas pequenas
- Métodos experimentais para a determinação da estrutura:
 - ❖ Cristalografia de raios X
 - ❖ Ressonância Magnética Nuclear (NMR)
 - ❖ Crio-Microscopia Electrónica
 - ❖ Métodos integrativos
 - ❖ Outros métodos (difração de neutrões, SAXS, etc)

Que informação temos disponível ?

(dados do Protein Data Bank em 1/02/2026)

Número de estruturas tridimensionais (coordenadas atómicas):

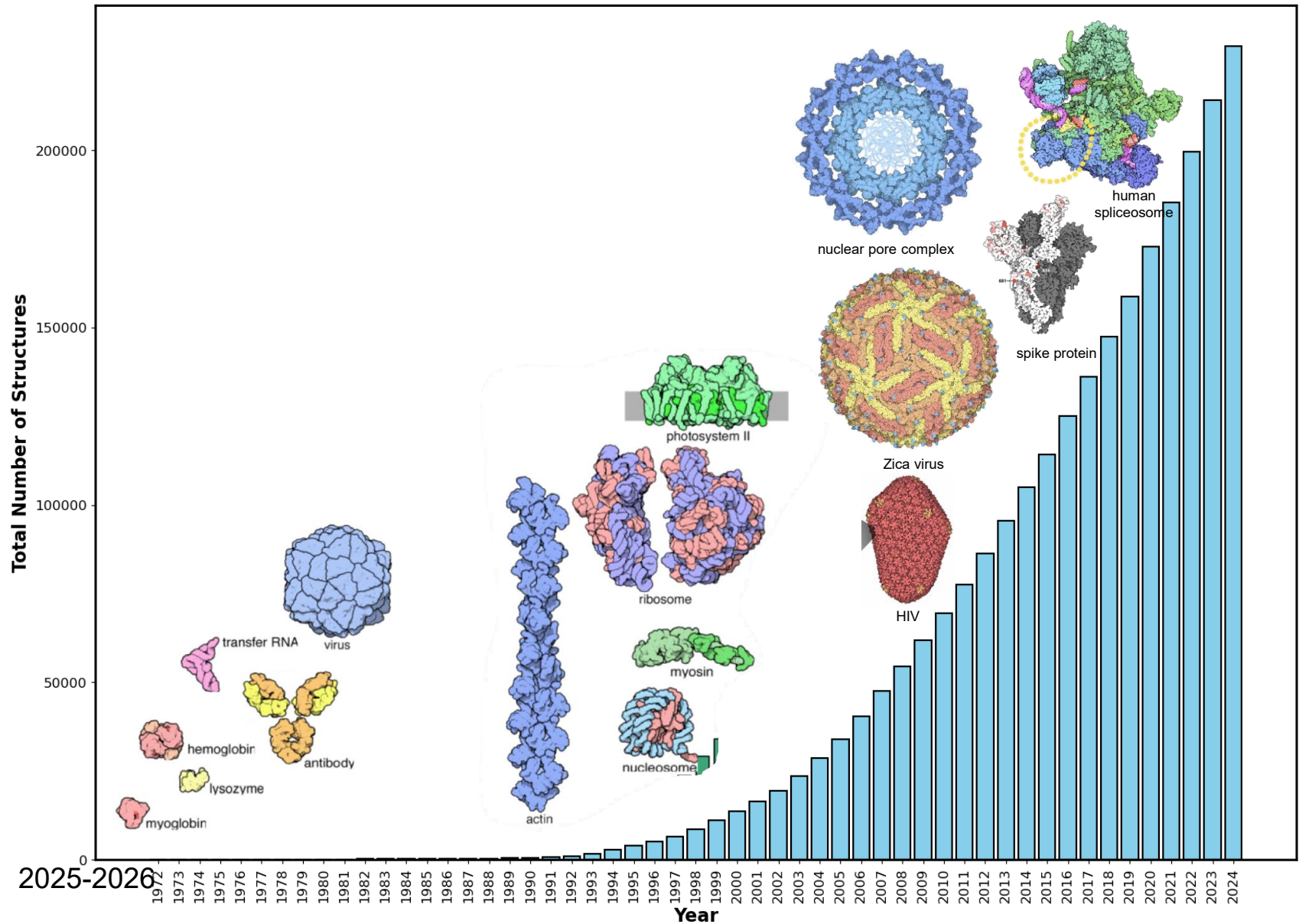
- Proteínas: 213858
- DNA/RNA: 4937
- Proteínas + Oligossacáridos: 13949
- Complexos ácido nucleico-proteína: 15714

Total: 248717 estruturas

Métodos experimentais de determinação da estrutura:

- Cristalografia de raios X: 201437
- Microscopia eletrónica: 31830
- NMR em solução aquosa: 14694
- Métodos integrativos: 382
- Outros: 347

Progresso na determinação das estruturas



Access Computed Structure Models (CSMs) of all available model organisms. [Learn more](#)

Welcome

Deposit

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Visualize

Analyze

Download

Learn

RCSB Protein Data Bank (RCSB PDB) enables breakthroughs in science and education by providing access and tools for exploration, visualization, and analysis of:

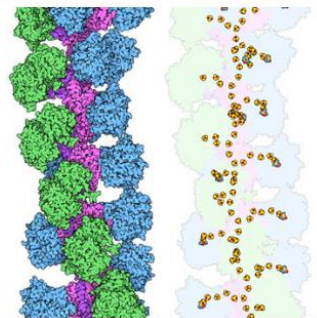
- Experimentally-determined 3D structures from the **Protein Data Bank (PDB)** archive
- Computed Structure Models (CSM)** from AlphaFold DB and ModelArchive

These data can be explored in context of external annotations providing a structural view of biology.

Explore NEW Features

PDB-101 Training Resources

February Molecule of the Month



Nanowires

Latest Entries

As of Tue Jan 30 2024



8GLK

The Type 9 Secretion System dGldL peak II, NucA substrate bound complex

Features & Highlights

Register for the February 5 RCSB.org Office Hour
Have questions about how to use RCSB.org? Join us for a virtual Office Hour.

Notice: NGL Viewer Deprecation
As of June 28, 2024, the NGL molecular viewer will be removed from RCSB.org. Users are encouraged to use Mol* for 3D visualization.

Take the CSM User Survey and Win
Please take this brief survey about Computed Structure Models at RCSB.org to be entered into a drawing for a set of Bound Playing Cards.

News

February 4 is World Cancer Day
PDB structures reveal how cell growth is normally controlled, and how cancer cells circumvent these essential controls
» 02/01/2024

Register for VIZBi 2024 (March 13–15)
Learn the latest data visualization methods and techniques
» 01/30/2024

Register for the February 13 Mol* Webinar
Join RCSB PDB to learn how to Visualize Biomolecular structures with Mol*: From Atoms to Movies
» 01/19/2024

Publications

PDB at a Glance 66,758 Structures of Human Sequences 17,179 Nucleic Acid Containing Structures
CSM at a Glance 999,251 AlphaFoldDB 69,326 ModelArchive [More Statistics](#)

2025-2026

<http://www.rcsb.org>

Princípios que regem a estrutura das biomoléculas

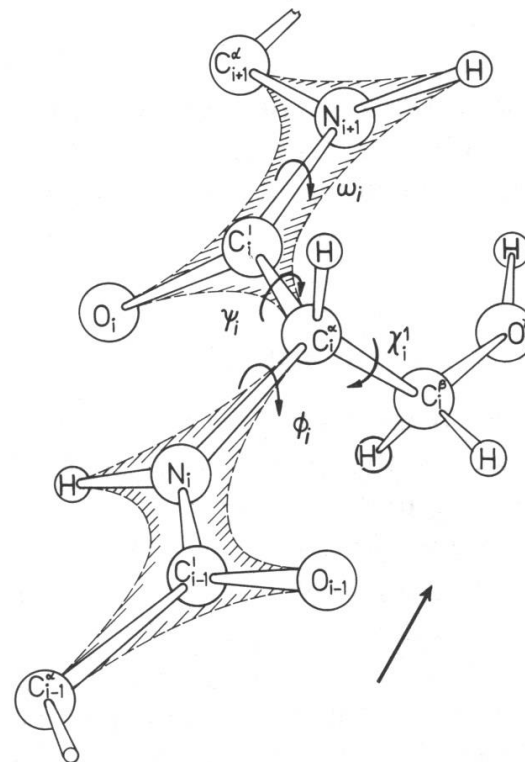
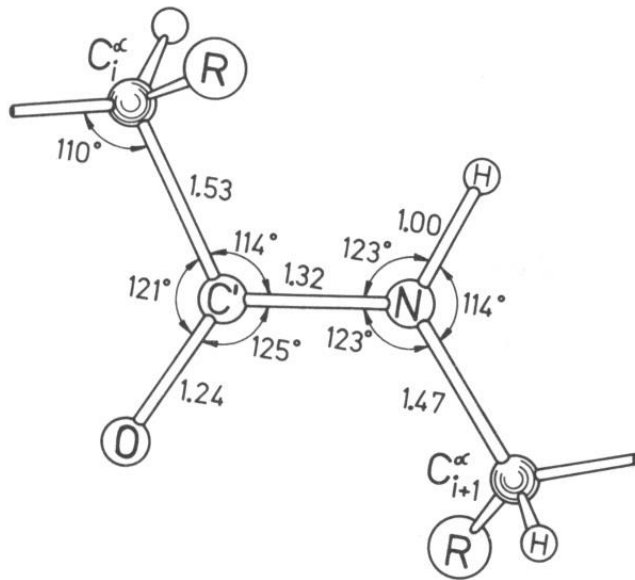
“Perhaps the most remarkable features of the molecule are its *complexity* and *lack of symmetry*. The arrangement seems to be almost totally lacking the kind of regularities which one instinctively anticipates and it is more complicated than has been predicted by any theory of protein structure”

J.C. Kendrew *et al.*, 1958

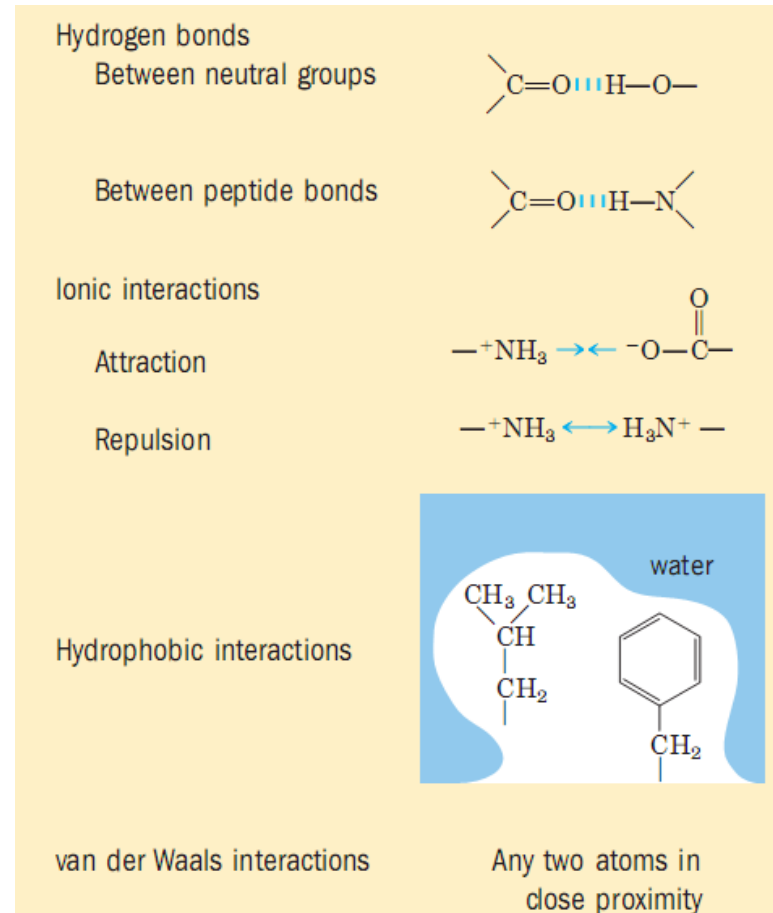


- As macromoléculas biológicas parecem, numa primeira análise, distanciar-se dos princípios simples de geometria e simetria que sabemos reger a estrutura das moléculas pequenas.

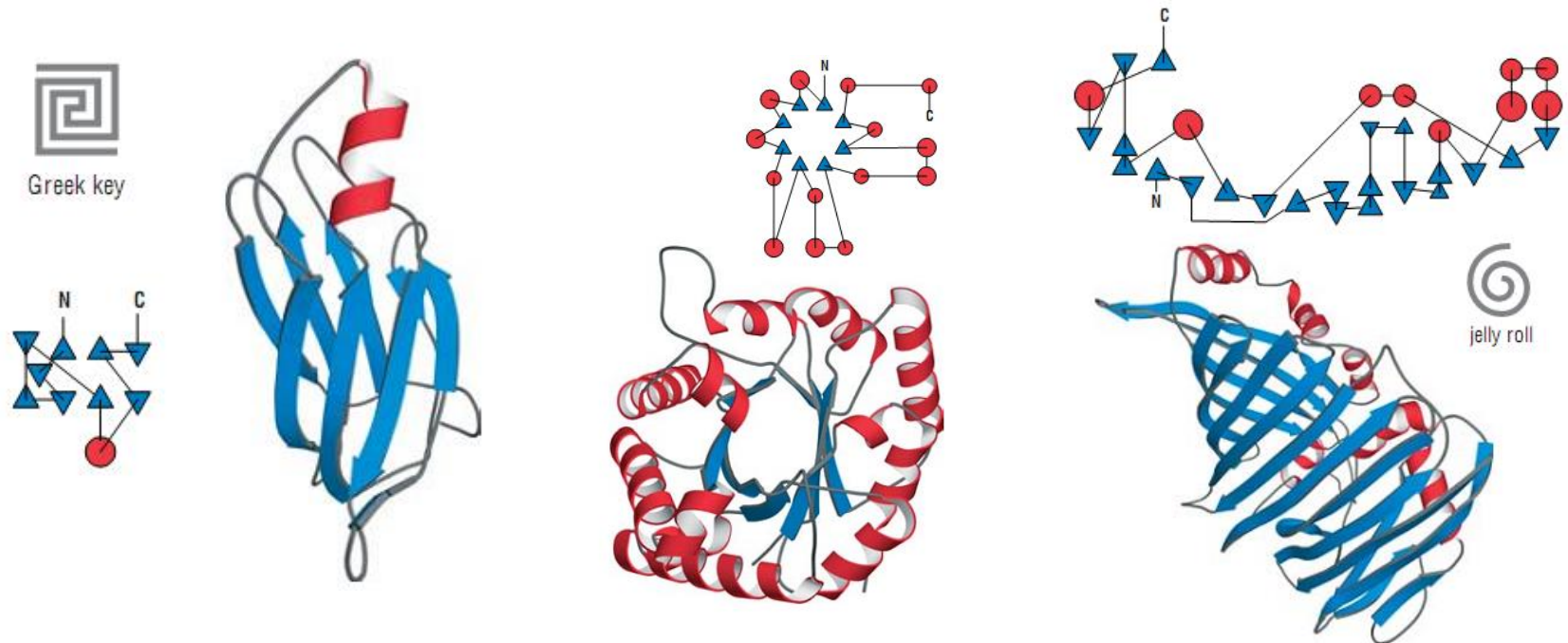
- Ligações covalentes, geometria molecular:



•Interações não-covalentes:

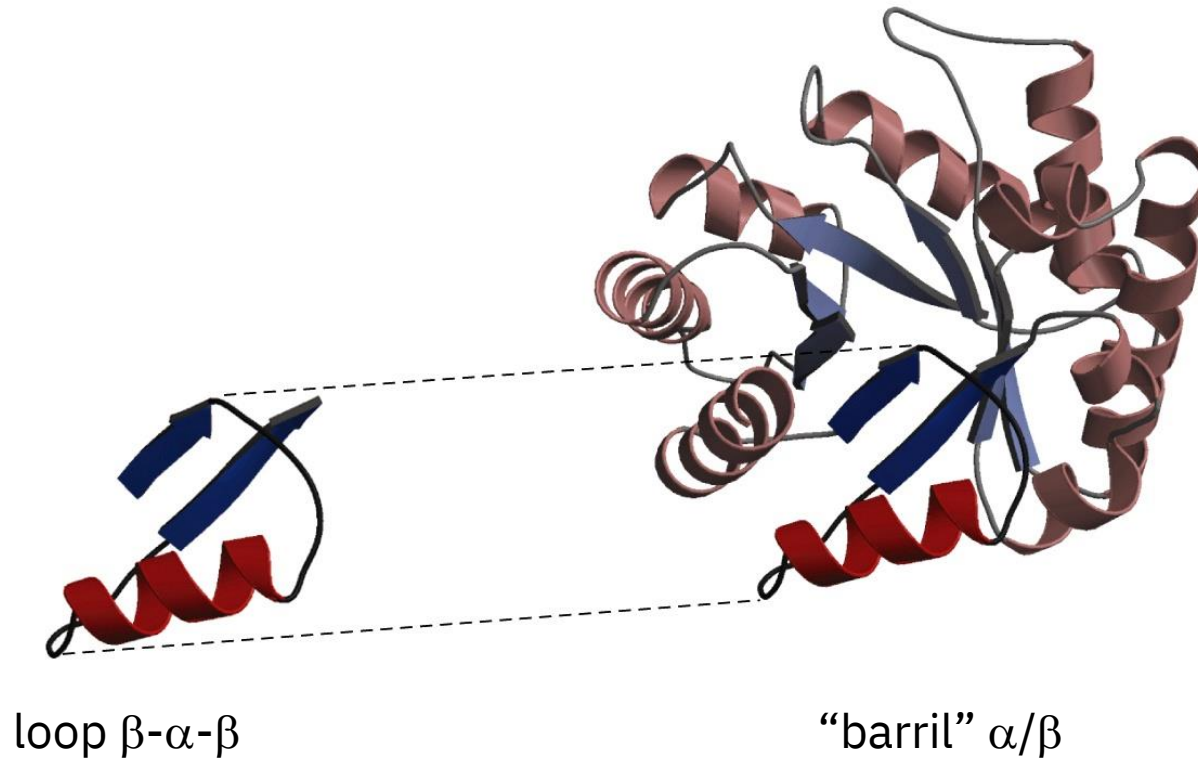


- Princípios arquitetónicos



Recorrência de padrões estruturais na arquitectura das biomoléculas.

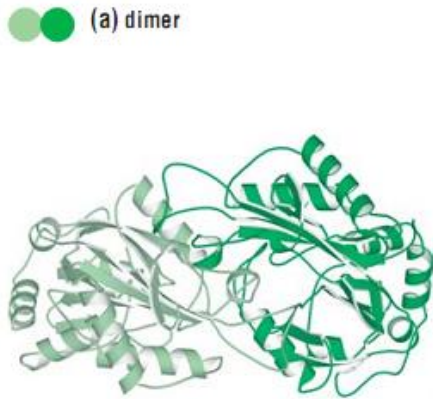
- Princípios arquitetónicos (cont.):



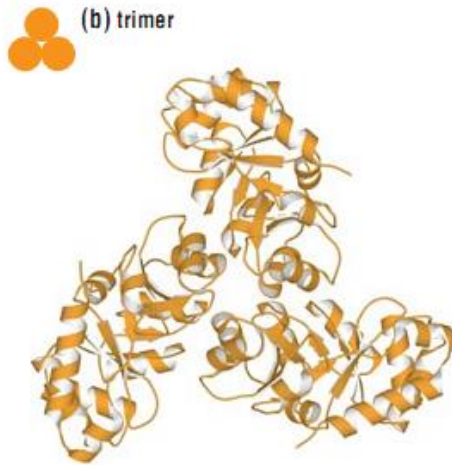
Formação de estruturas a partir da associação de unidades estruturais

- Oligomerização:

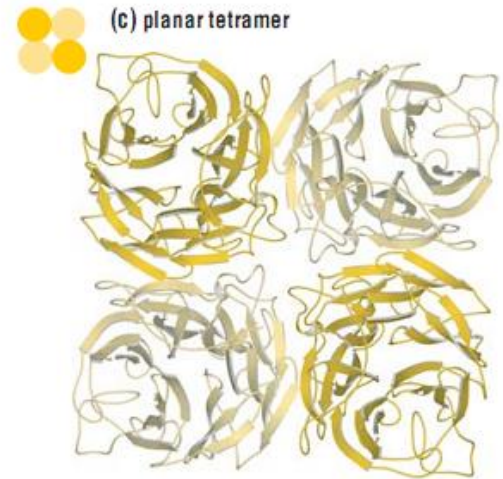
(a) dimer



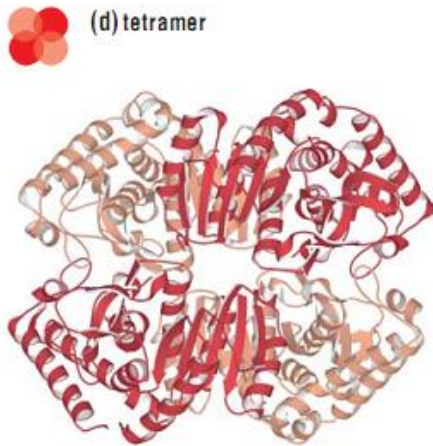
(b) trimer



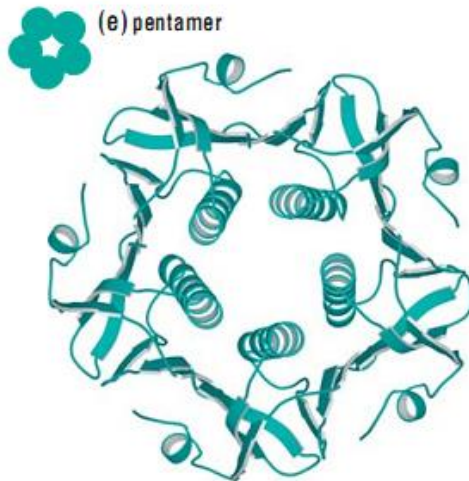
(c) planar tetramer



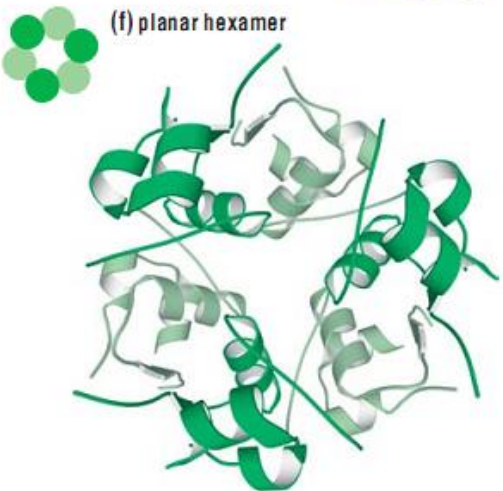
(d) tetramer



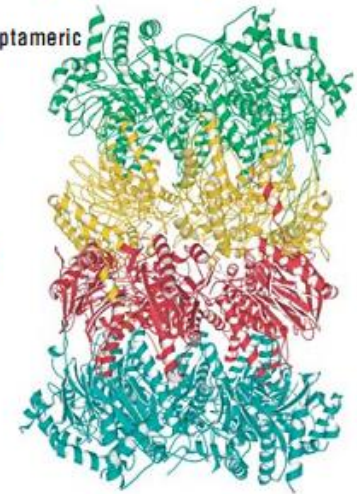
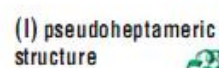
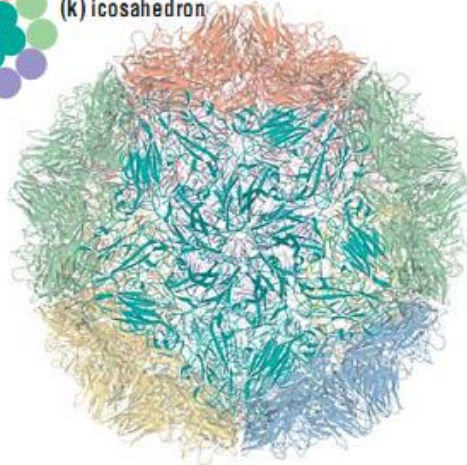
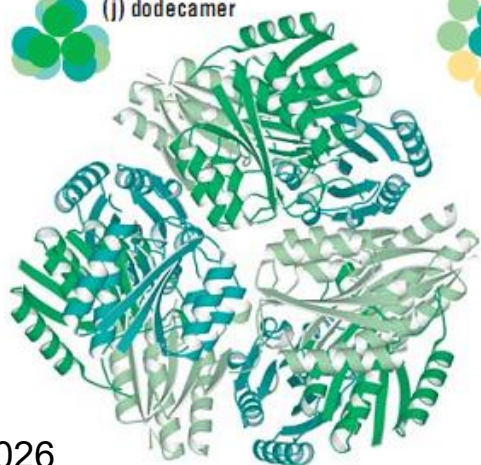
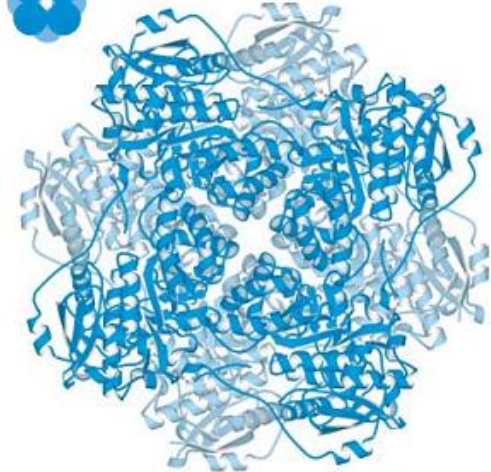
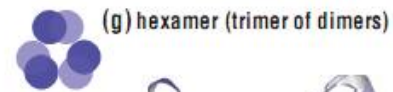
(e) pentamer



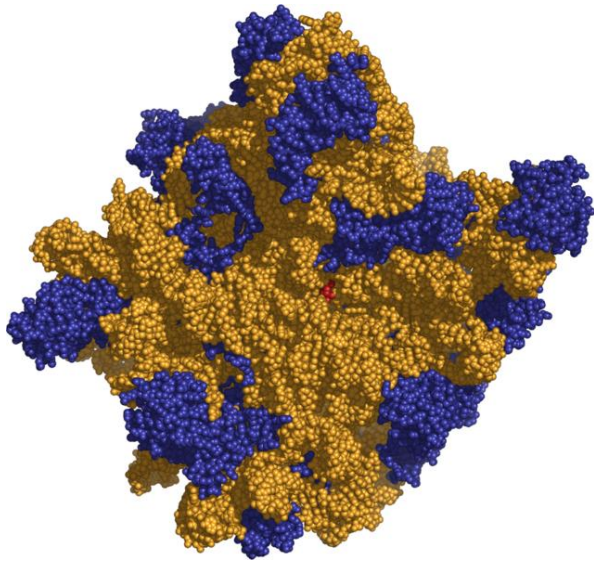
(f) planar hexamer



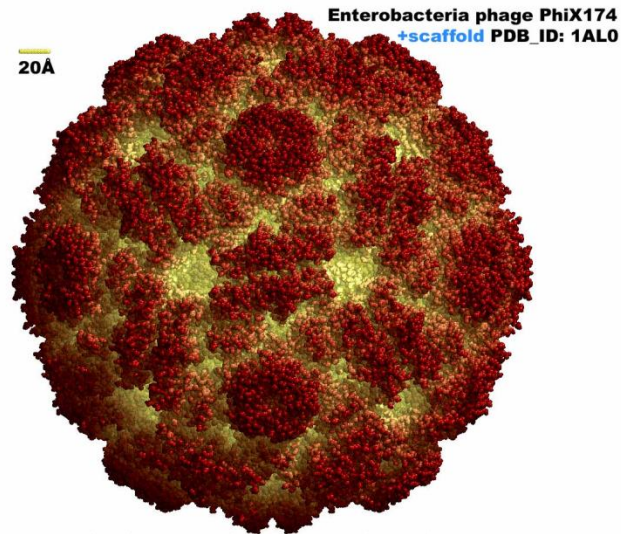
- Oligomerização(cont.):



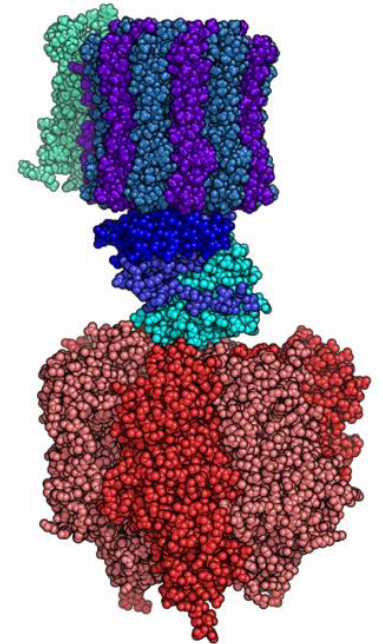
- Formação de estruturas supramacromoleculares



Ribossoma



Vírus

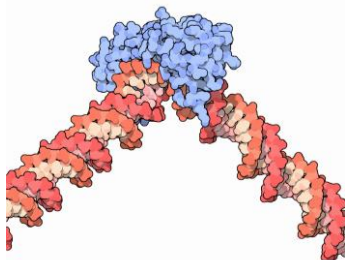
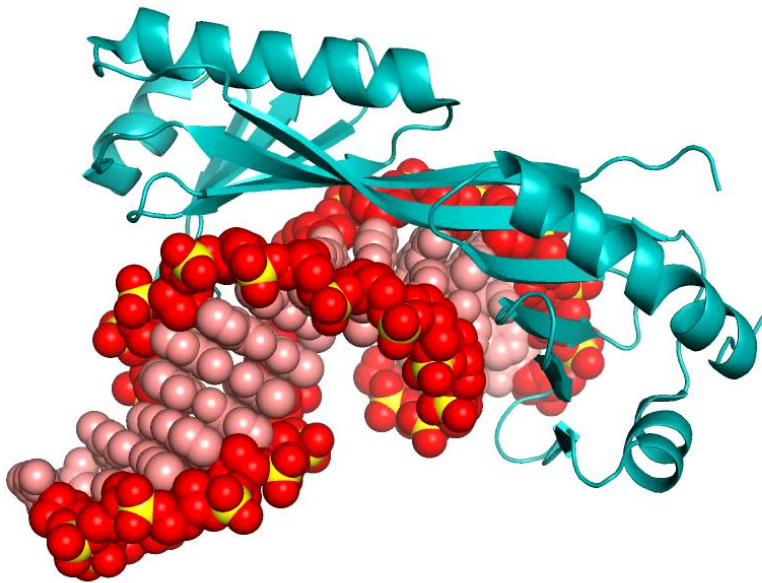


ATPase

Função

Associação a ligandos

TATA-binding protein

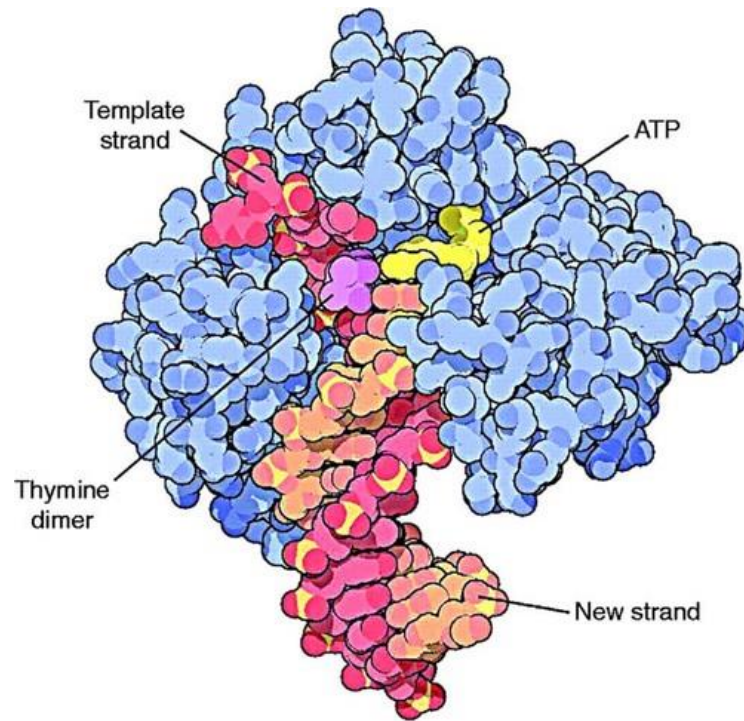


Myoglobin

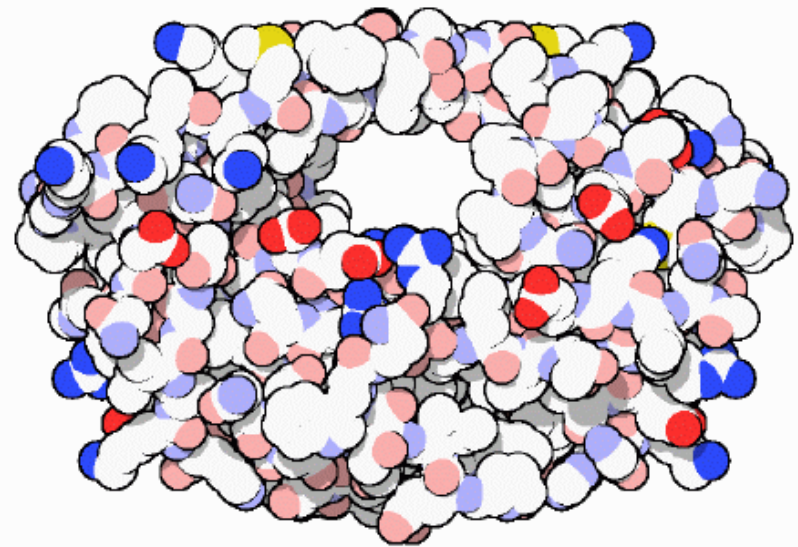


Catálise

DNA polymerase

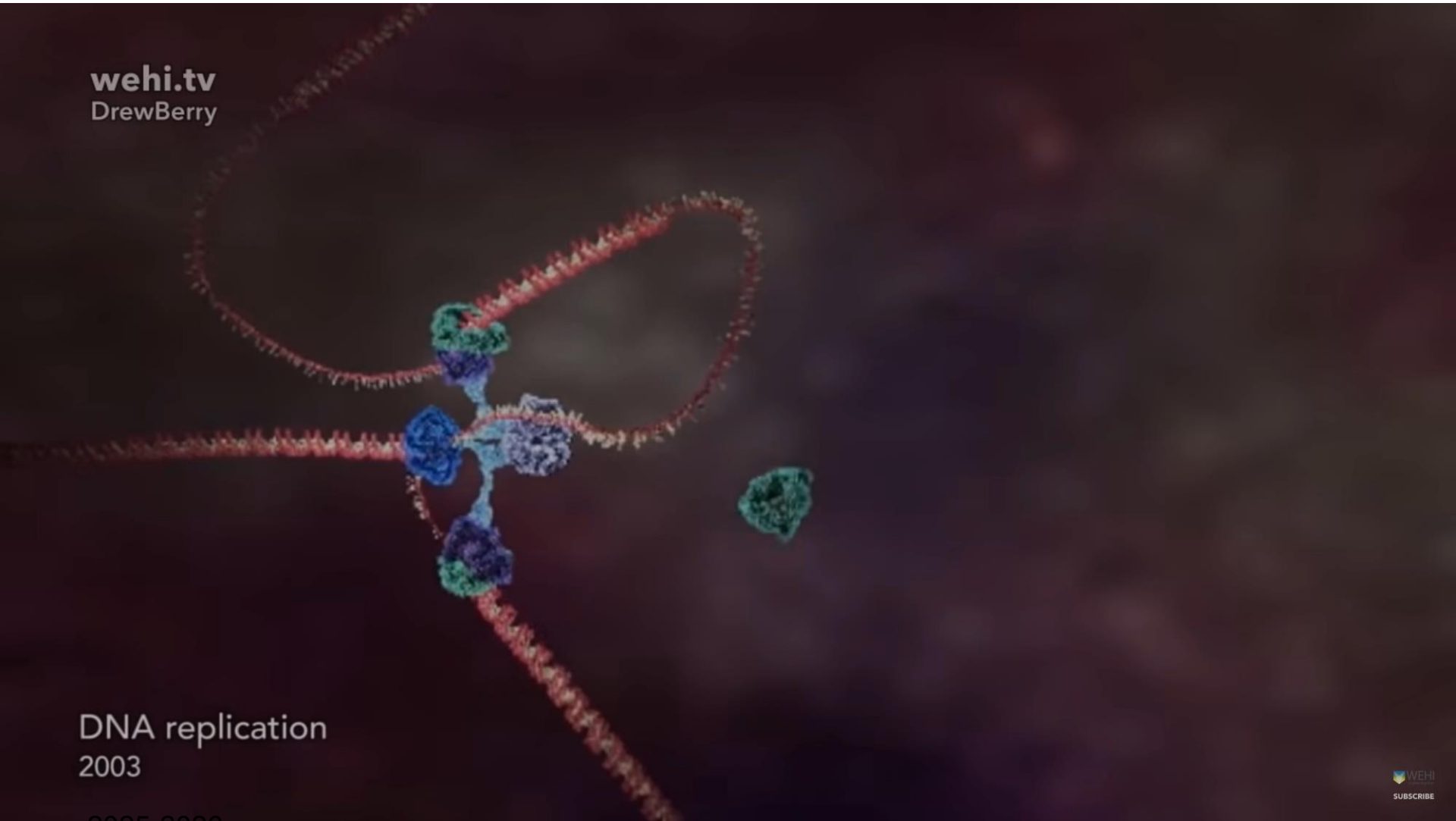


HIV protease



“Fotocopiadora” molecular

wehi.tv
DrewBerry

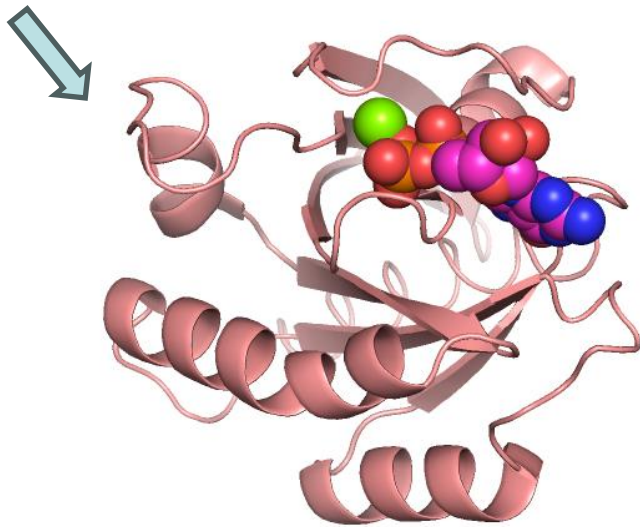


DNA replication
2003

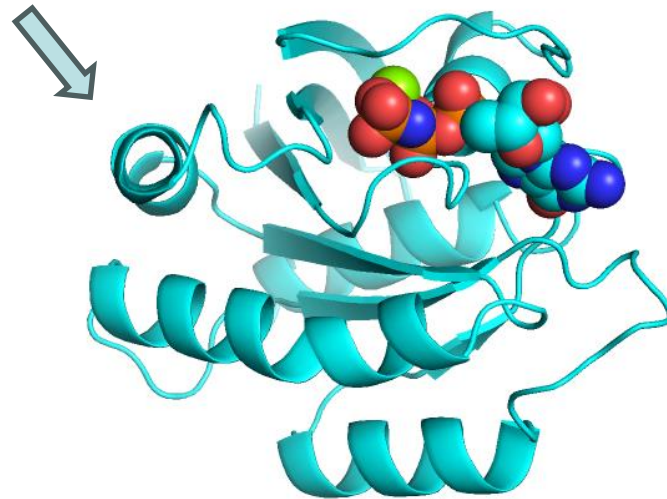
2025-2026

Sinalização

ras protein

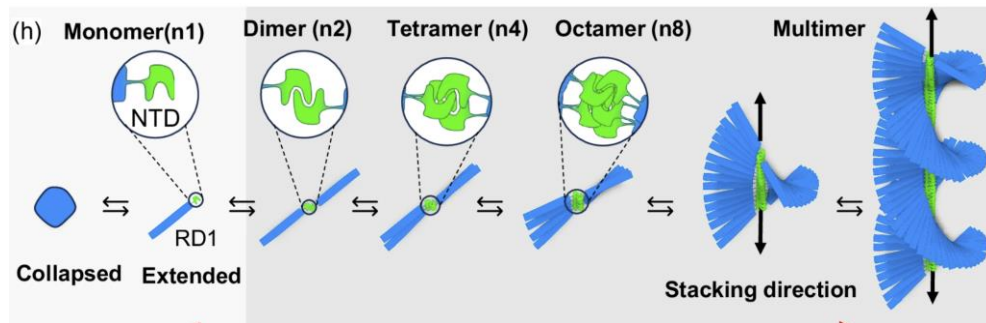
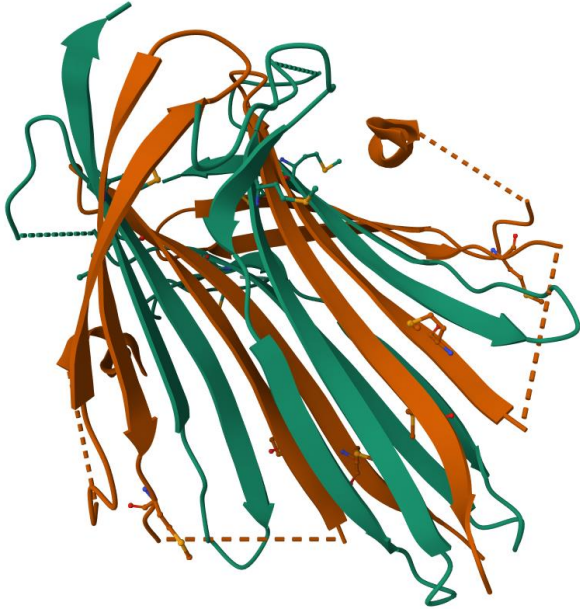


On



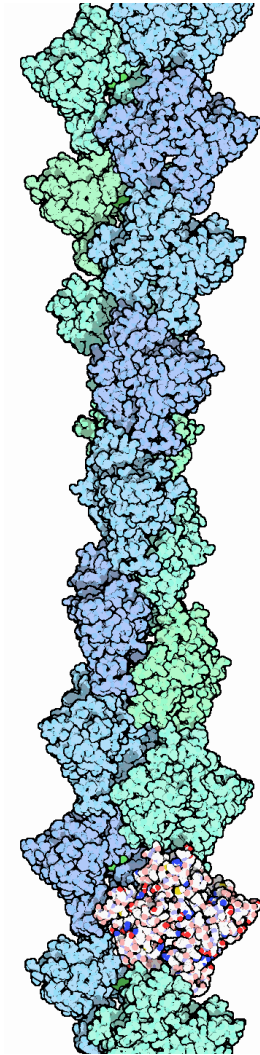
Off

Estrutura



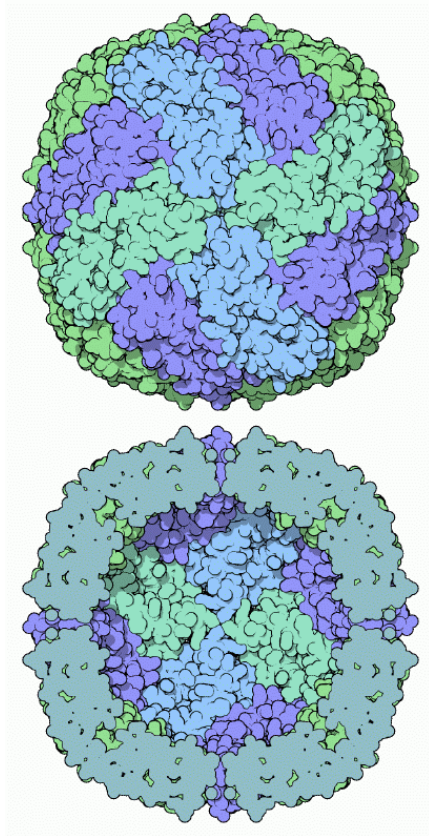
Silk fibroin

2025-2026

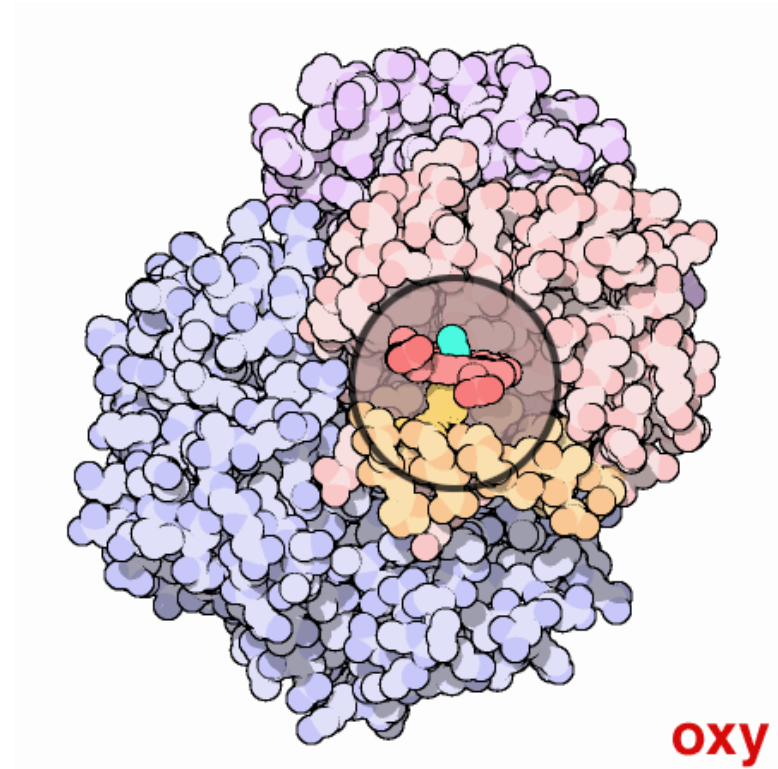


F-actin

Transporte

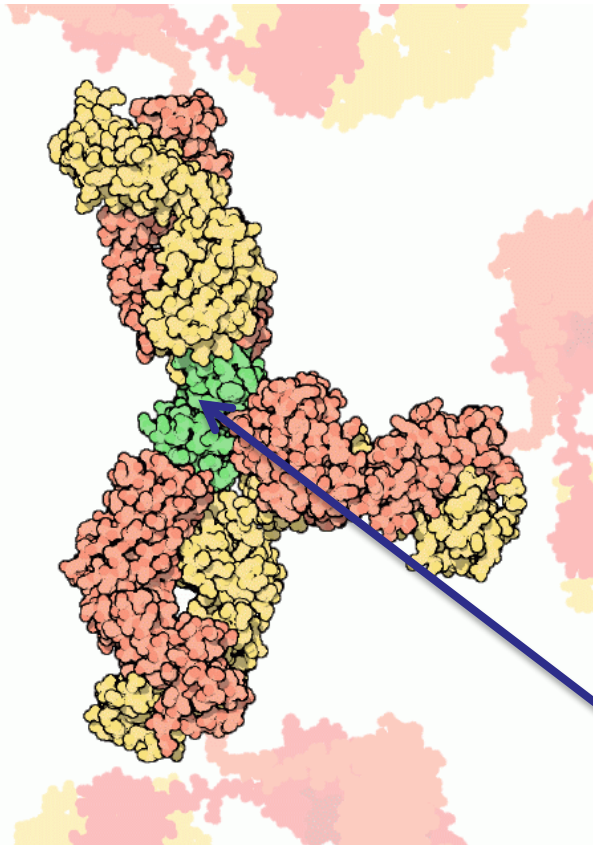


Ferritina (ferro)

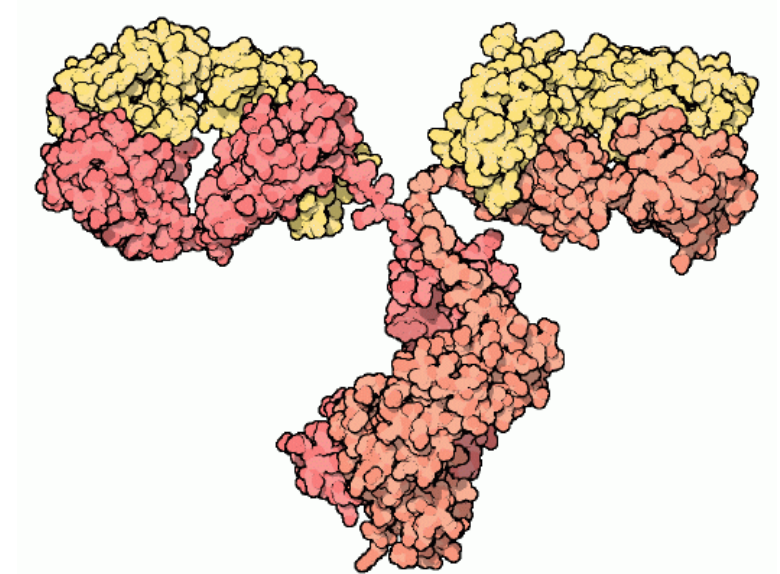


Hemoglobina (oxigénio)

Defesa

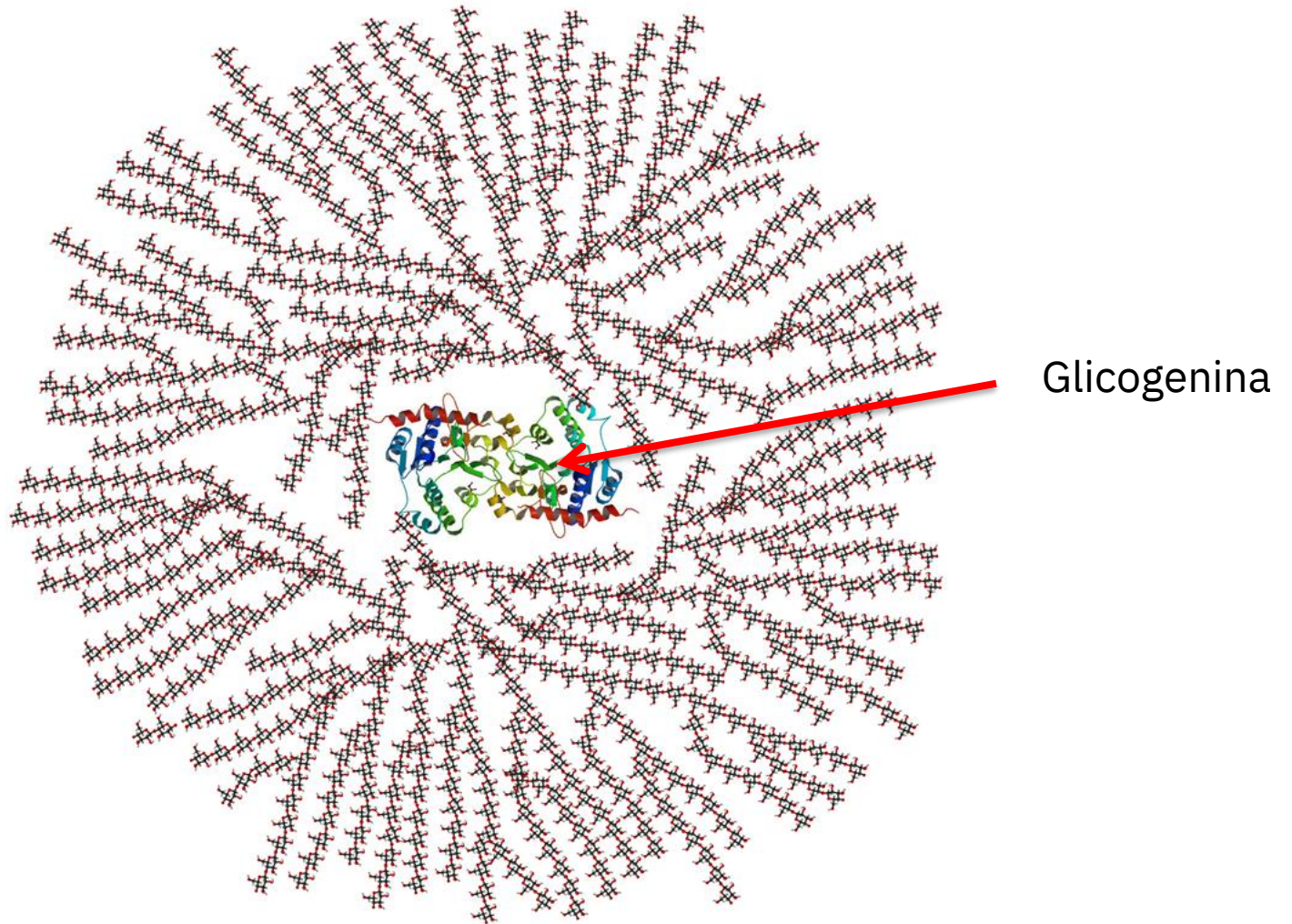


lisozima

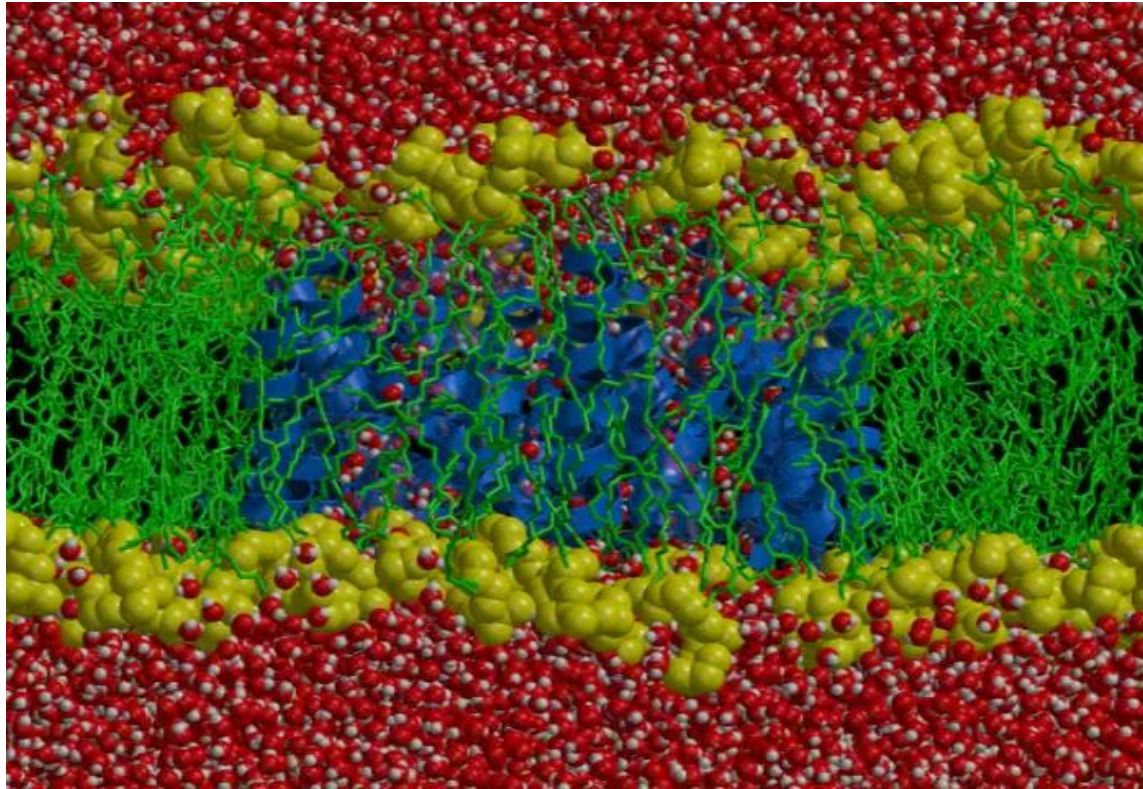


imunoglobulina

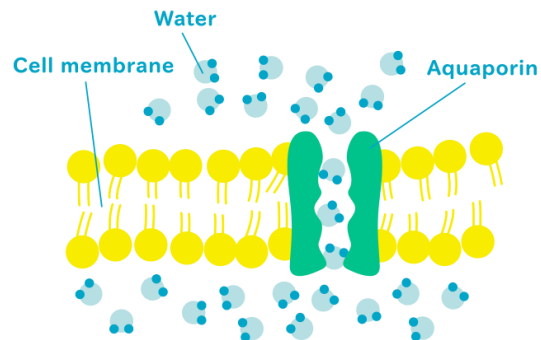
Armazenamento



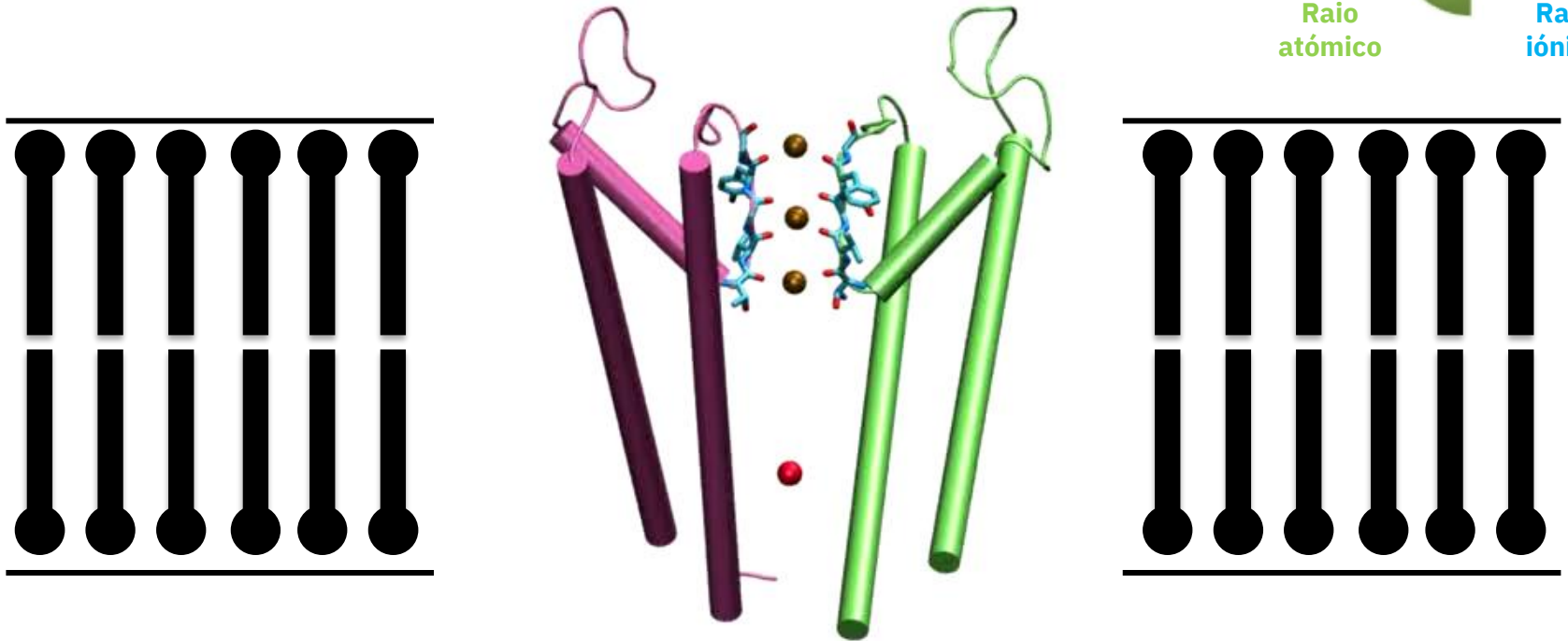
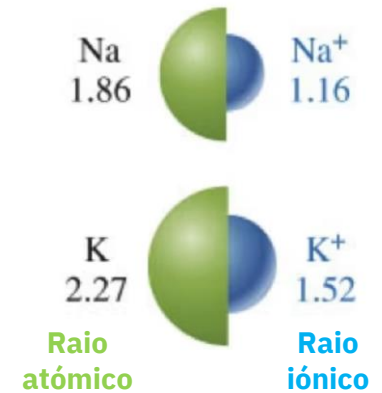
Compartimentação



Membrana+aquaporina



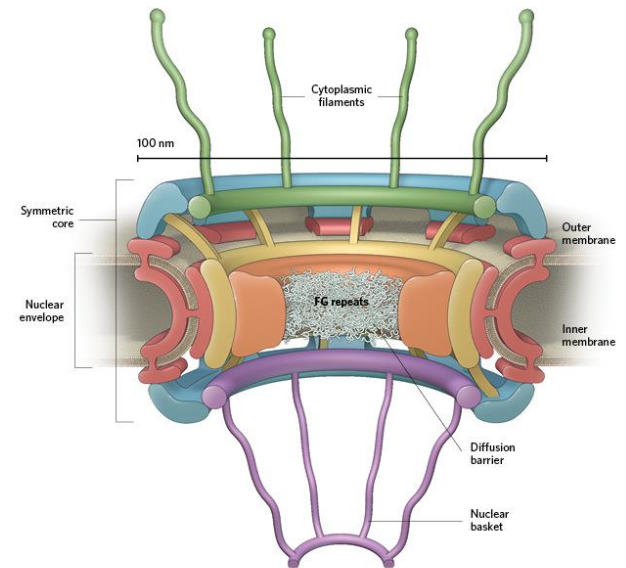
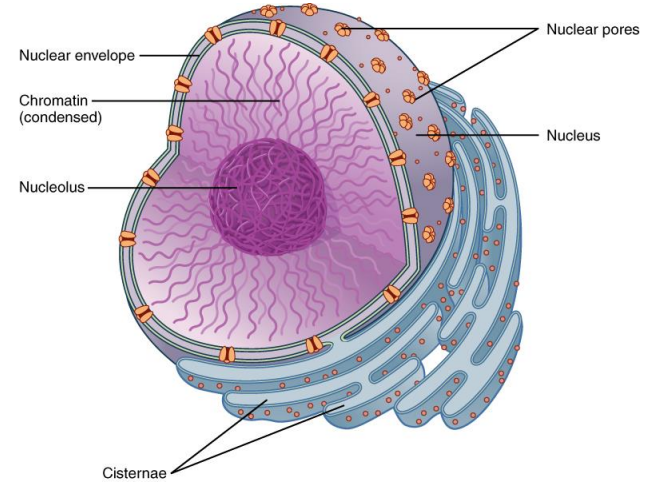
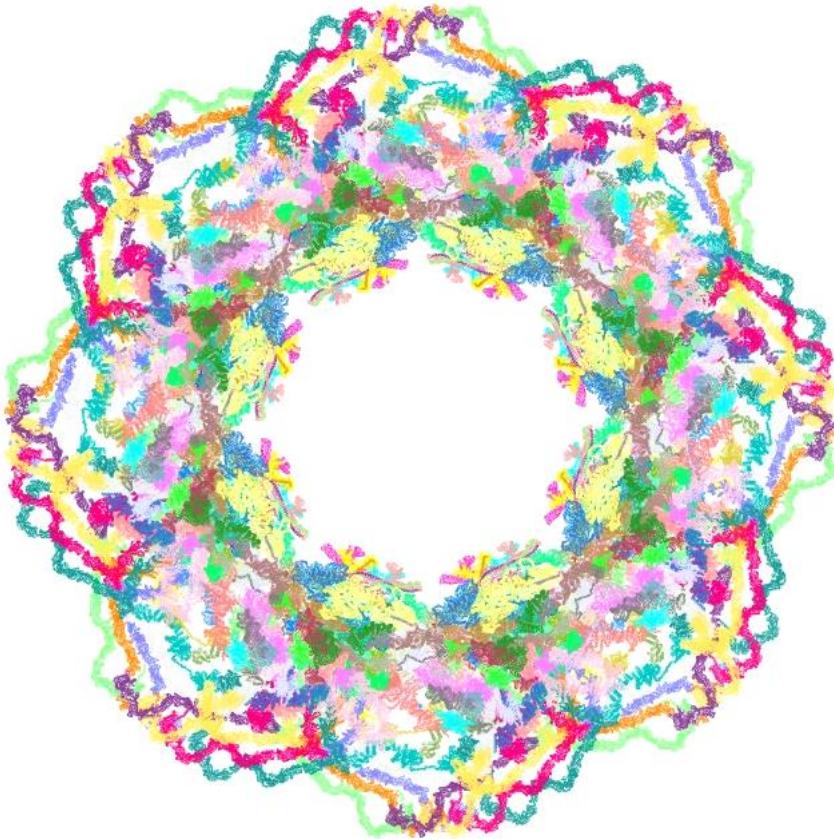
Permeabilidade e transporte (canal de potássio KV1.2)



Transporte passivo, **selectivo** para íons potássio, íons sódio não passam o canal (embora o **raio iônico** do potássio seja superior ao do sódio)!

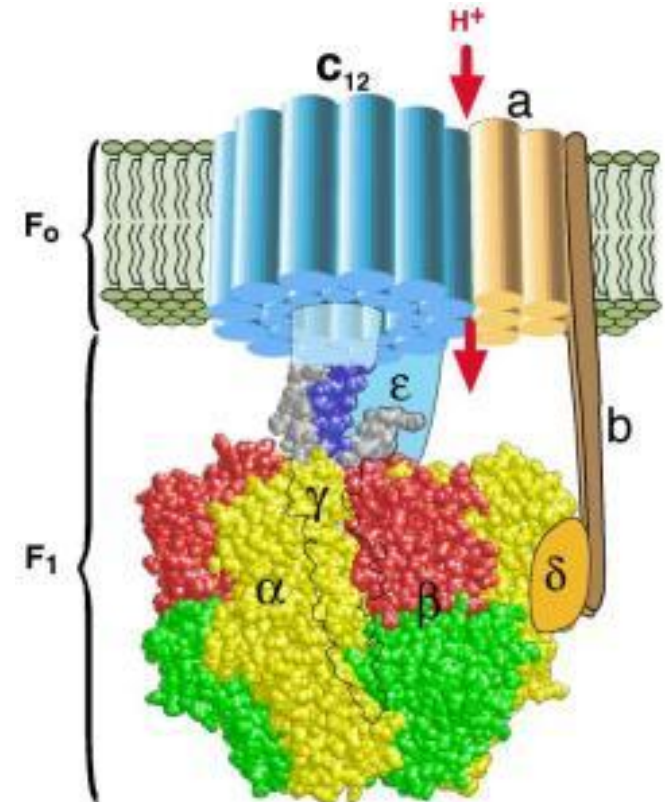
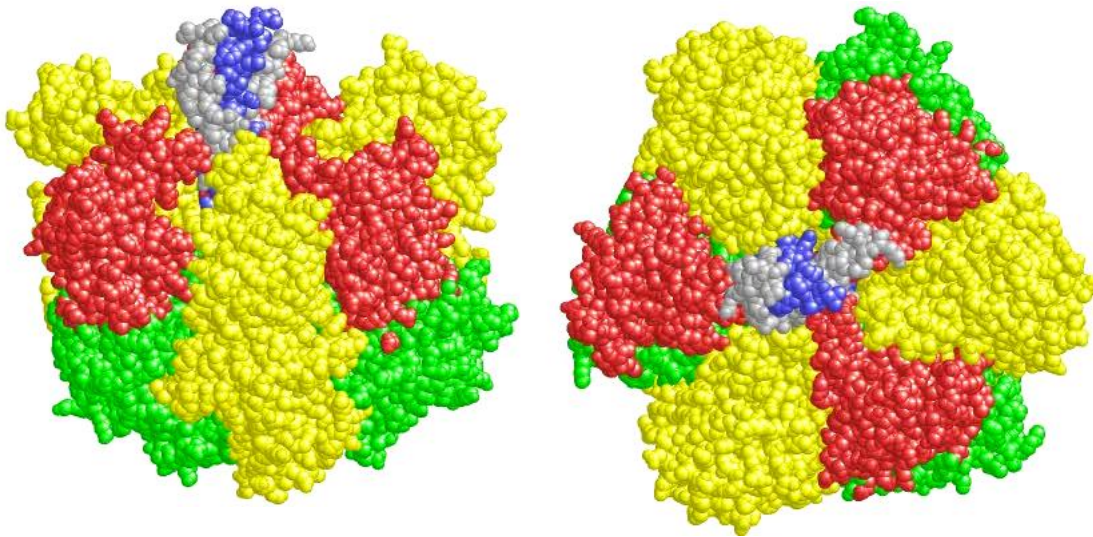
PDB: 2A79

Estrutura de um poro nuclear

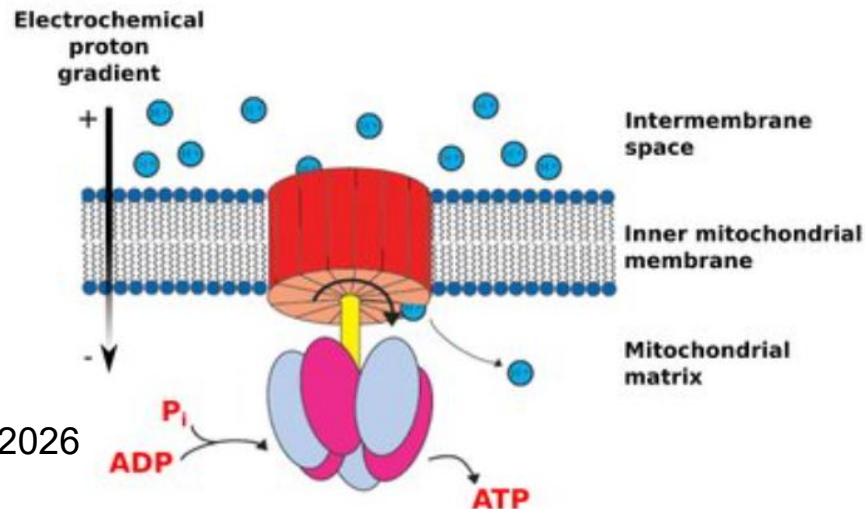


Conversão de energia

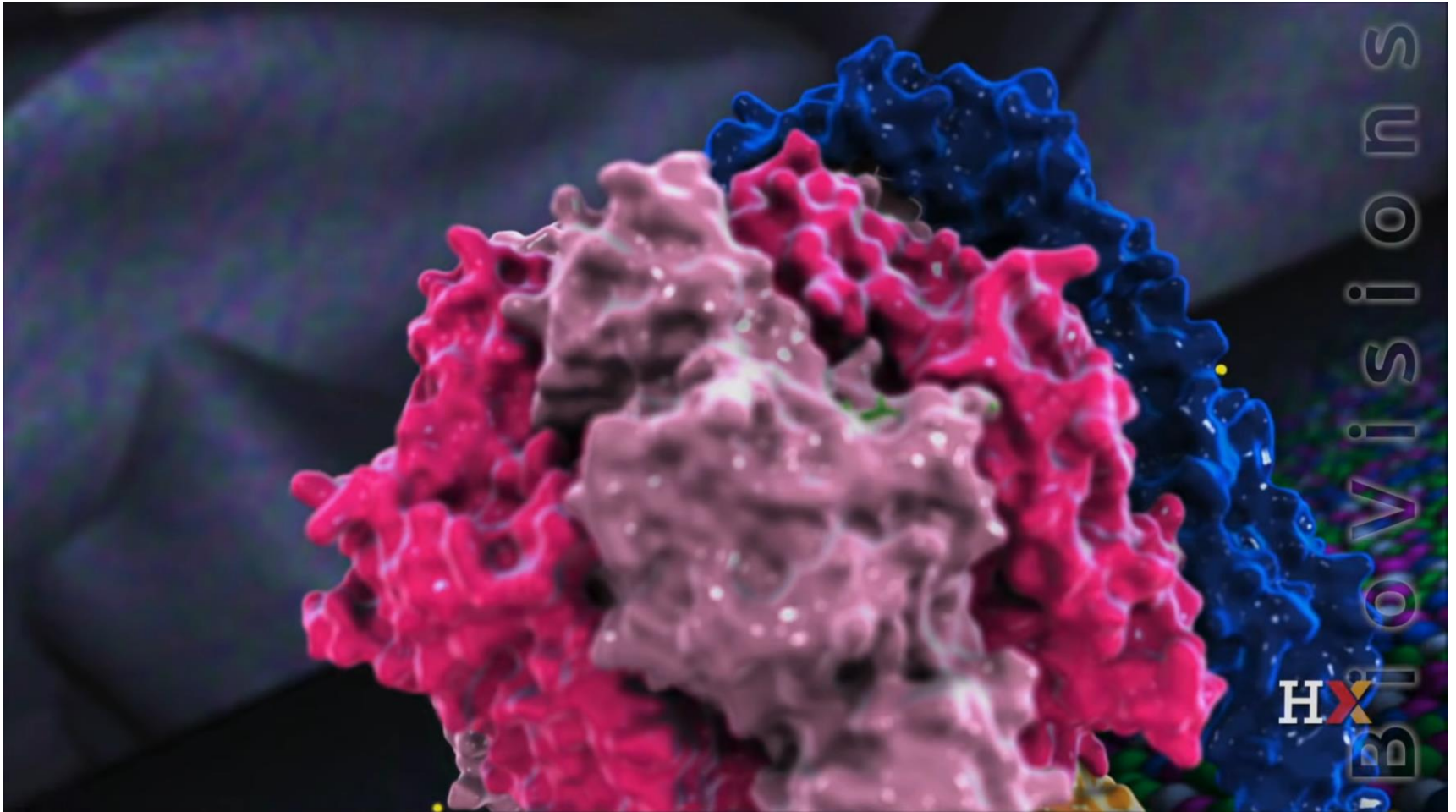
ATP sintase



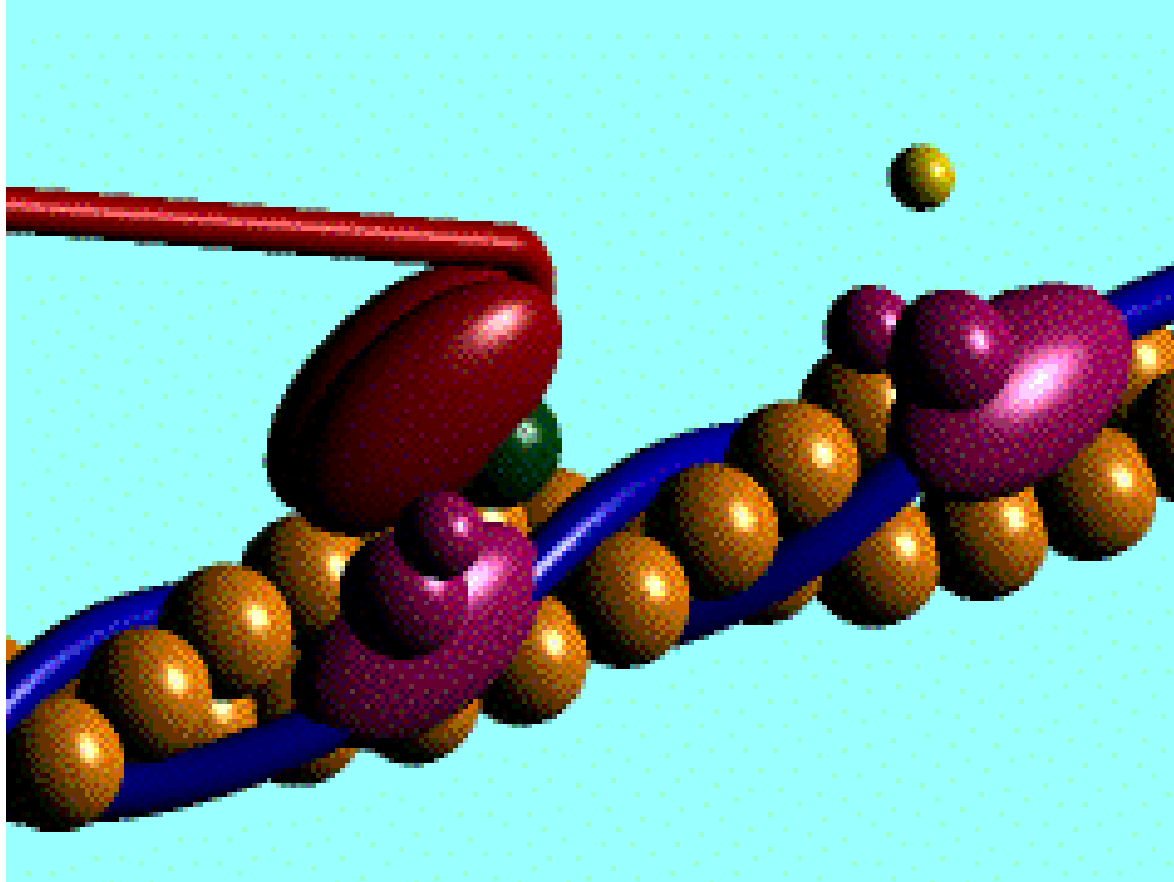
H. Wang and G. Oster (1998). Nature 396:279-282.



A molecular power generator

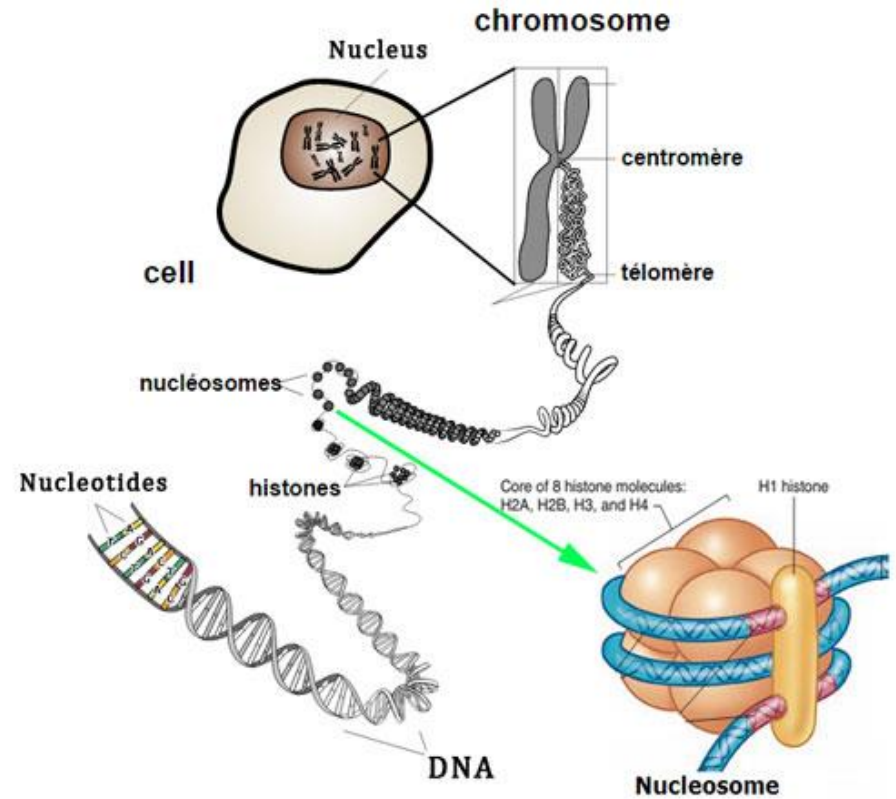
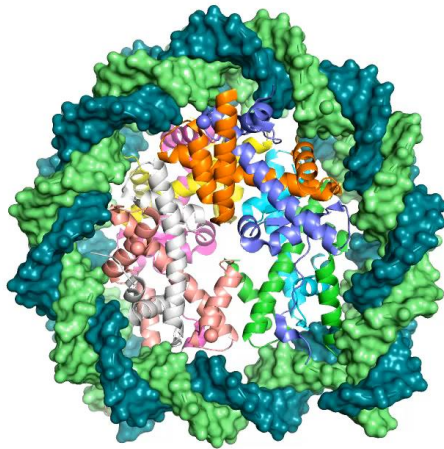
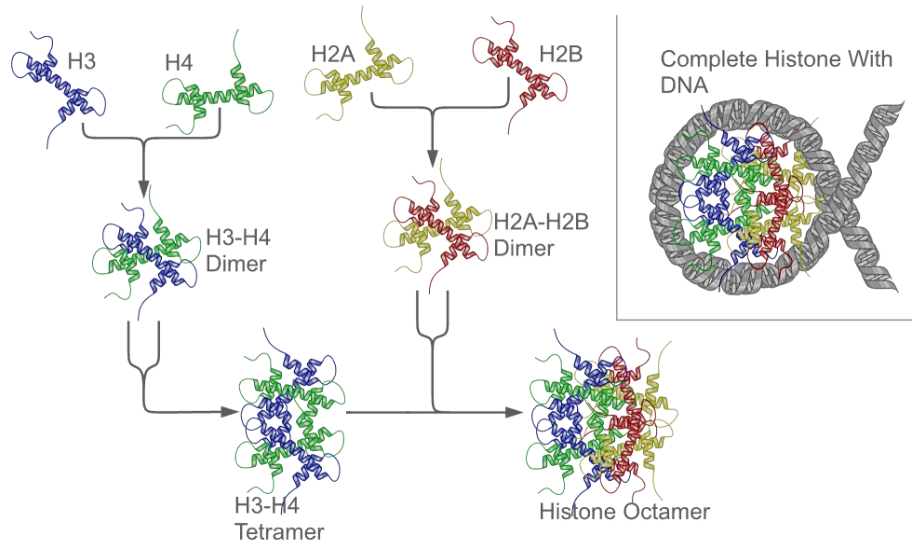


Motilidade

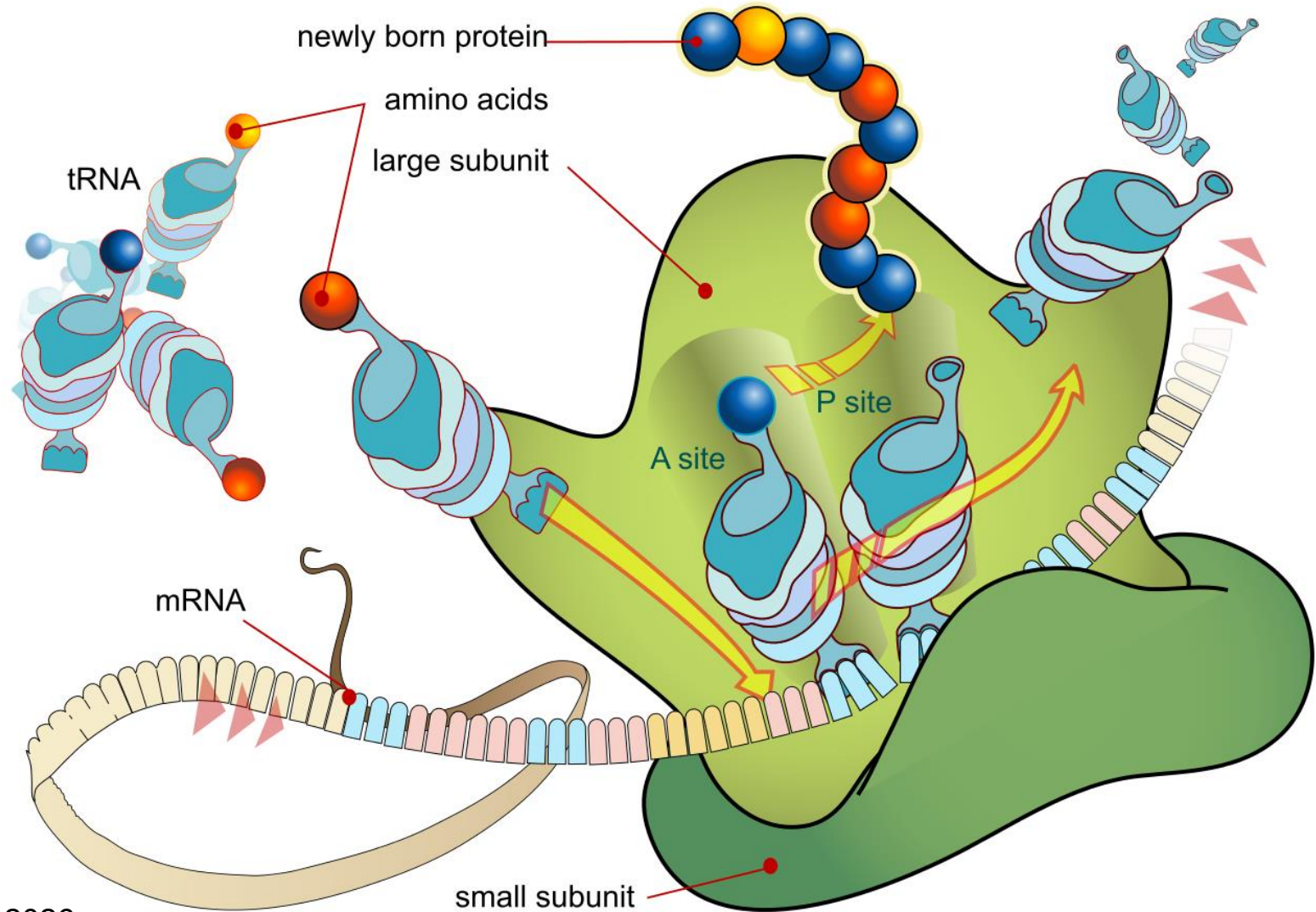


Actina+Miosina

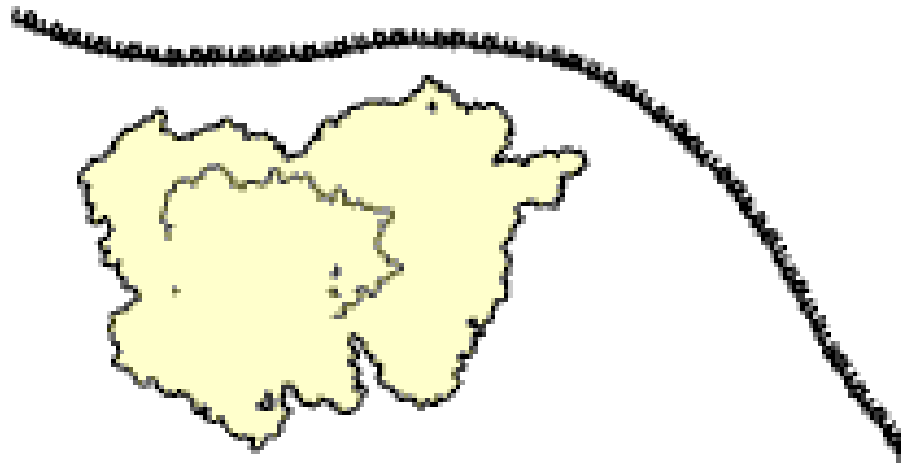
Informação



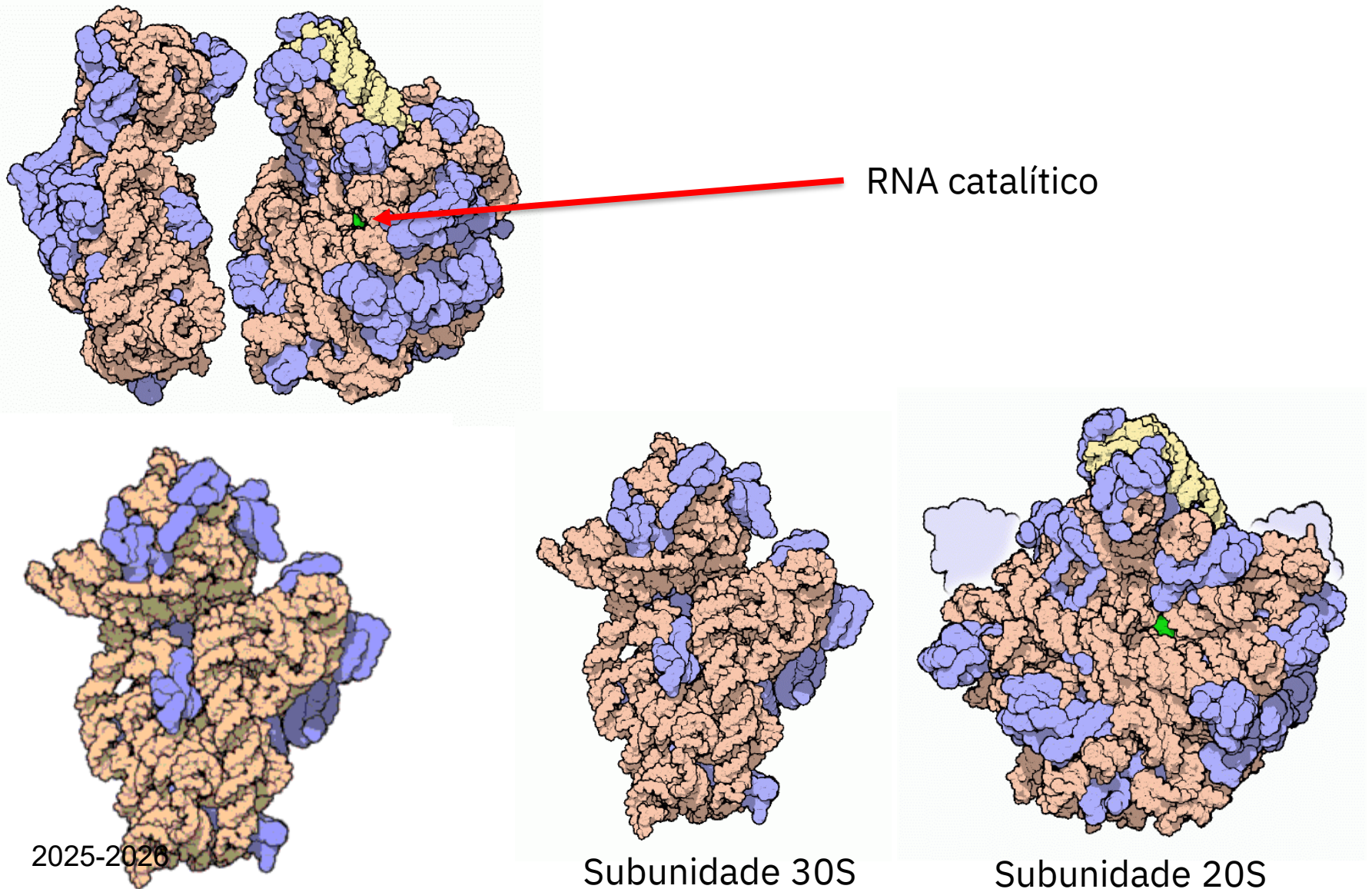
Síntese proteica



Síntese proteica (animação)

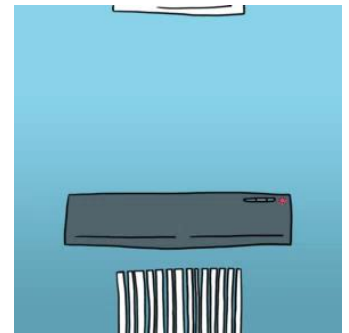
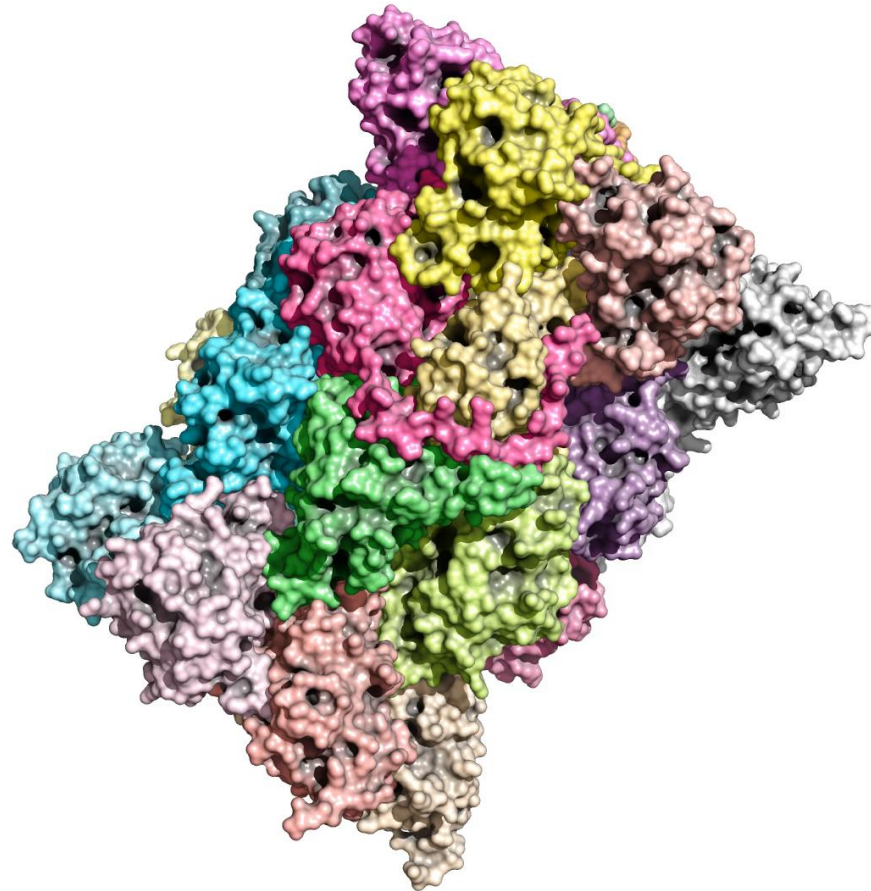


Estrutura do ribossoma bacteriano



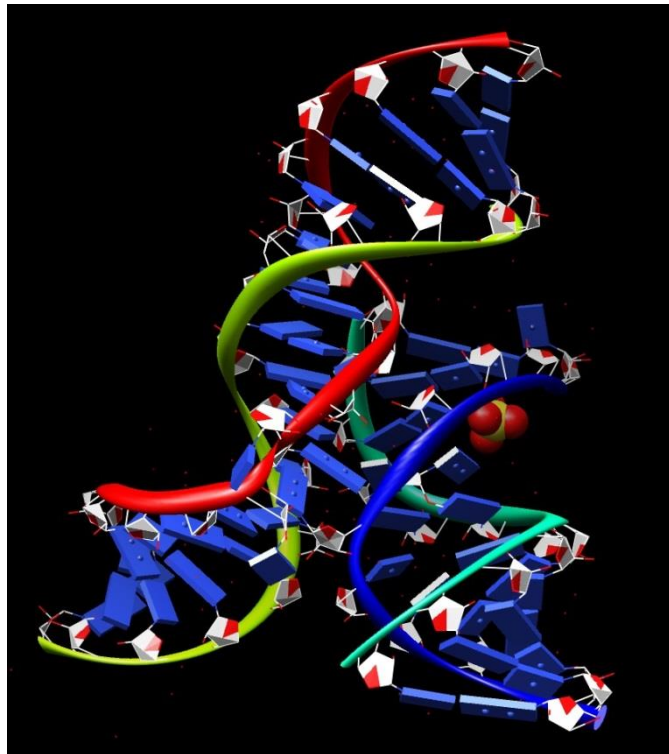
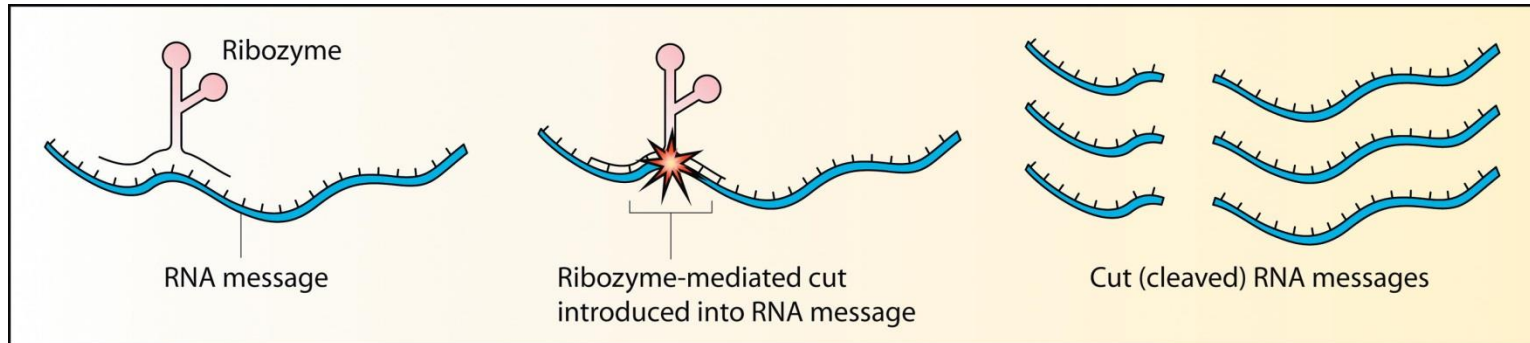
Reciclagem de proteínas

Proteassoma

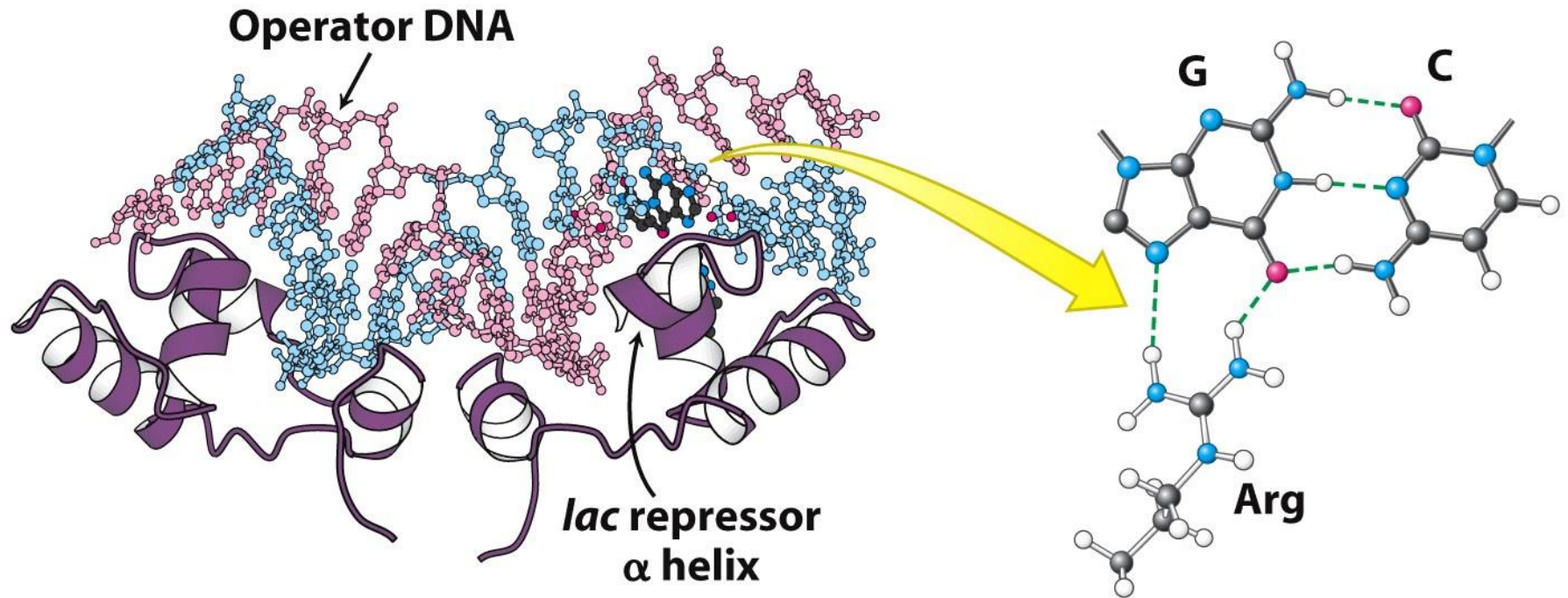


Assemblagem de máquinas moleculares

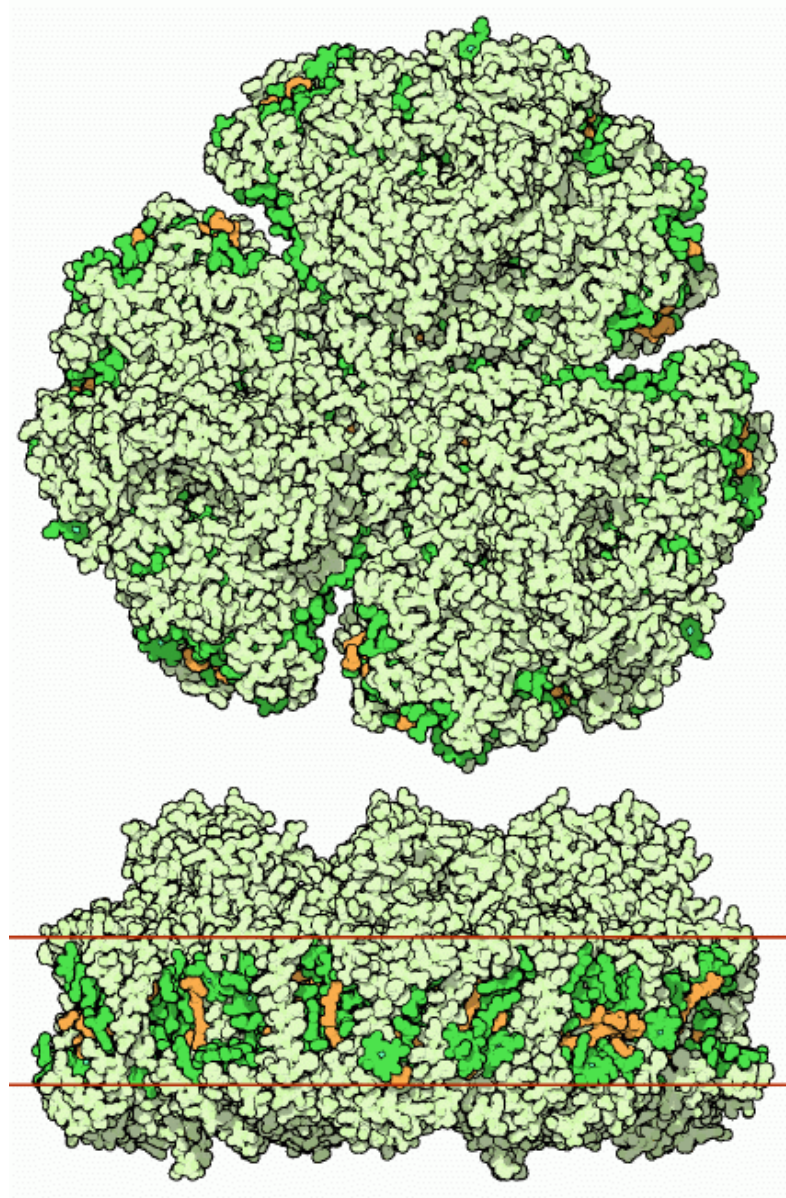
Ribozimas



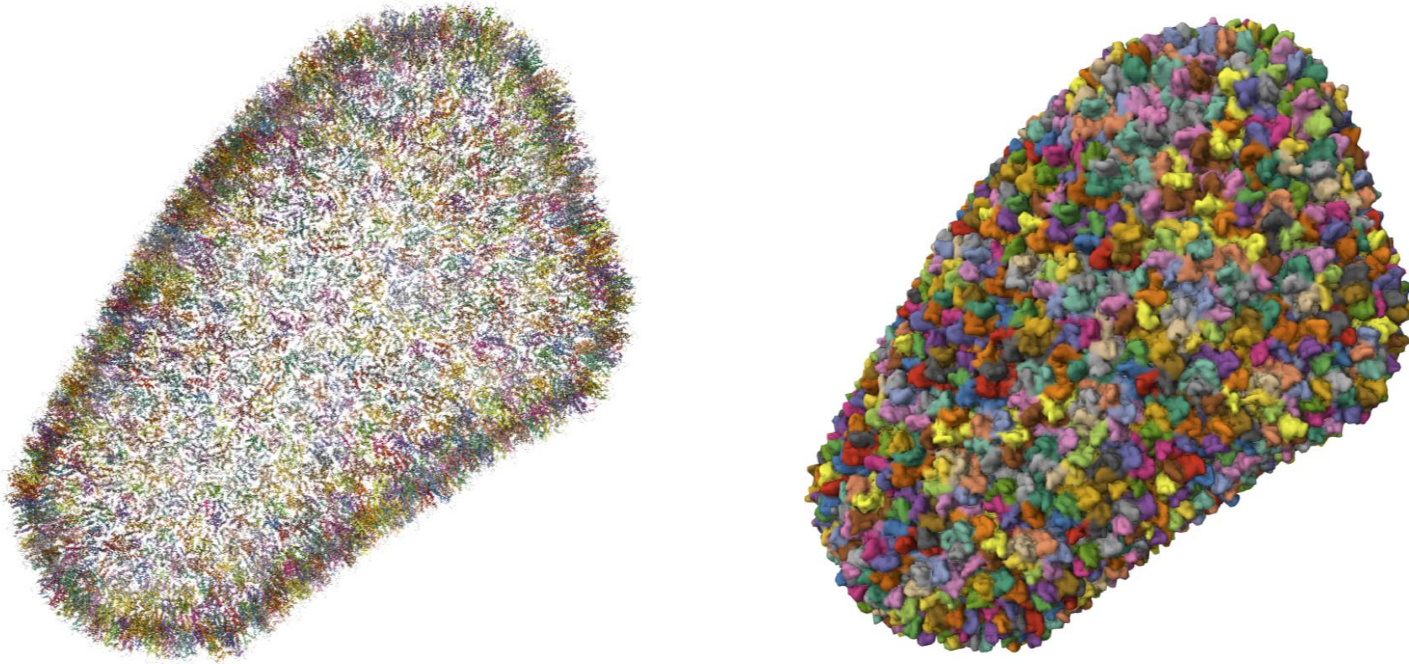
Regulação da transcrição



Fotossistema I

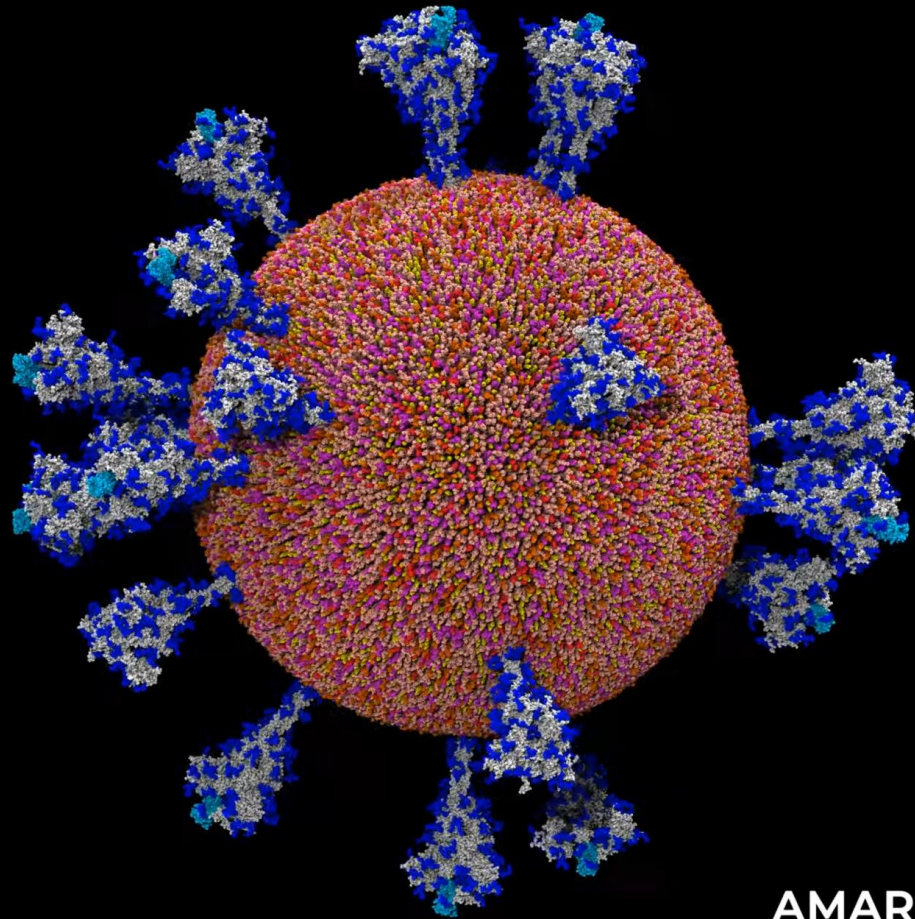


Cápside Viral (HIV1)



Estrutura de baixa resolução (8.0 Å) resolvida por cryo-EM

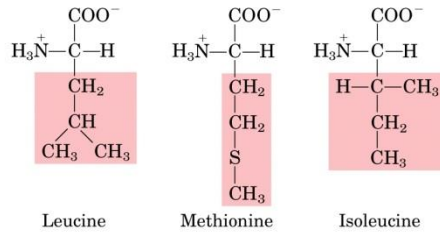
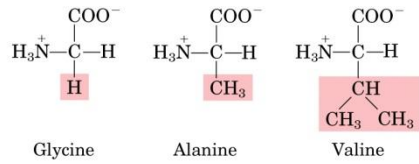
3.5M-atom model of SARS-CoV-2



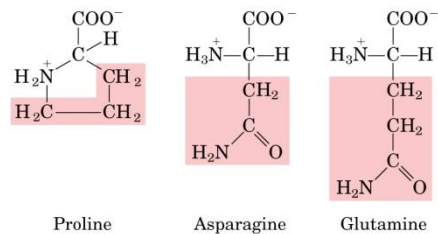
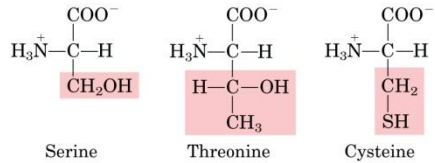
AMARO LAB - UCSD

Proteínas

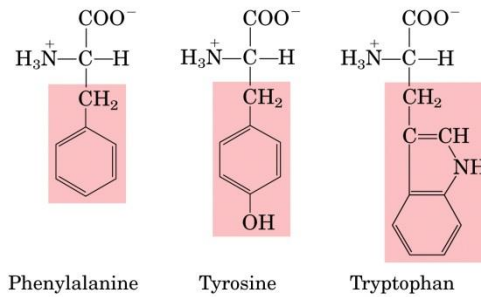
Nonpolar, aliphatic R groups



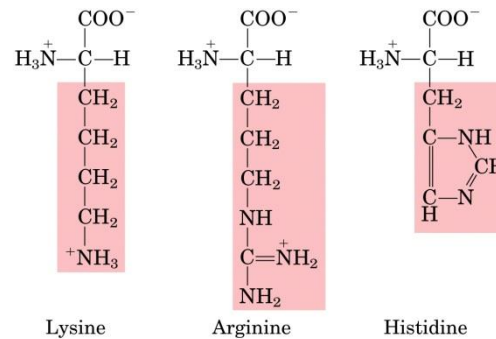
Polar, uncharged R groups



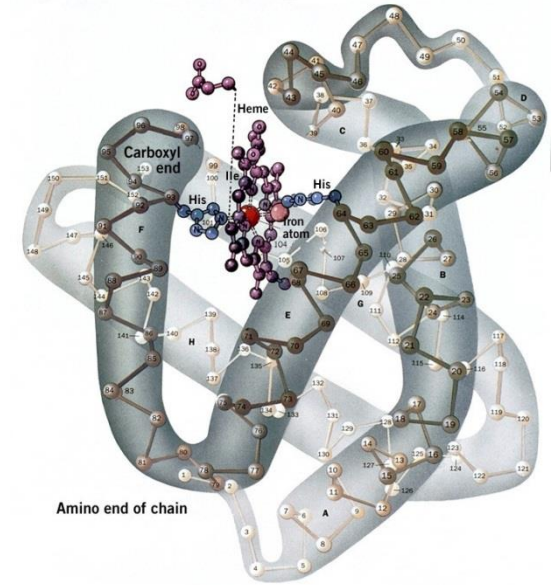
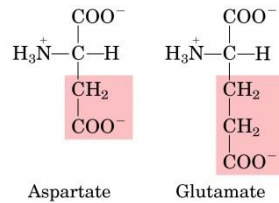
Aromatic R groups



Positively charged R groups



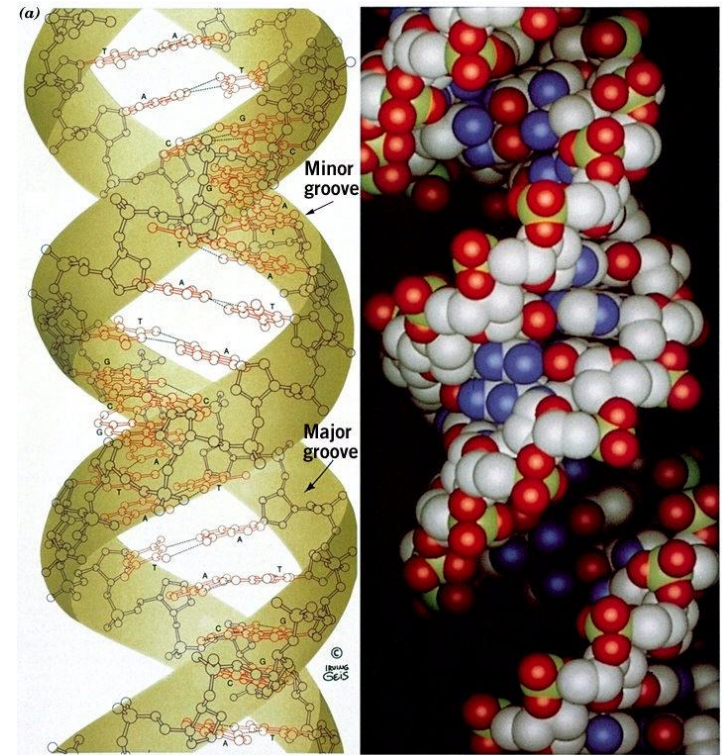
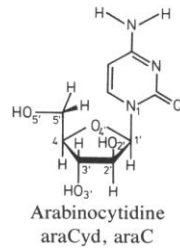
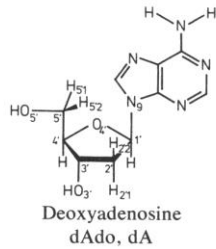
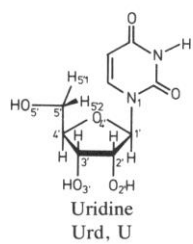
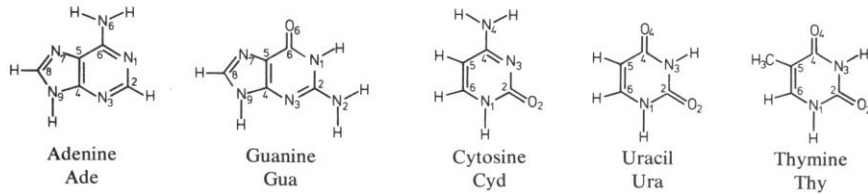
Negatively charged R groups



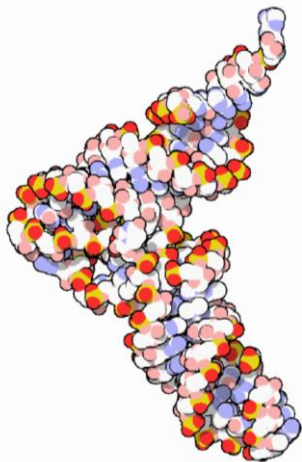
(a)

Mioglobina

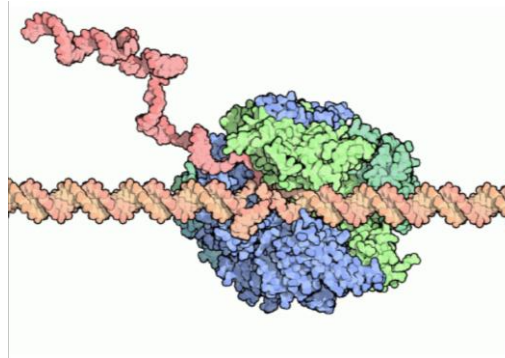
Ácidos nucleicos- blocos constituintes



B-Dna



RNA de transferência
2025-2026



RNA mensageiro